Decision-time admissibility in predictive process monitoring: measuring what causal feature selection costs, and what it buys

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Abstract

Predictive process monitoring reports strong discrimination on operational event logs, but much of that performance depends on attributes that are unavailable when the decision they inform is actually taken. We state a decision-time admissibility protocol, requiring every attribute to be observable at the moment of assignment and every group-level target encoding to be estimated from past cases only, and we measure what that discipline costs on the BPI Challenge 2014 incident log, 31 236 incidents from a banking IT service desk. Across four gradient-boosted learners, enforcing admissibility removes between 0.0955 and 0.1038 AUC, a quarter of all above-chance signal, and every 95 % bootstrap interval on that loss excludes zero. We then measure what admissibility buys. An admissible feature ladder rises from 0.5076 at three attributes to 0.7859 at seventeen without any inadmissible attribute. Modelling case resolution as a discrete-time hazard and deriving breach as a conditional survival ratio reaches 0.8017 AUC at intake against 0.7399 for a direct classifier, while remaining structurally unable to observe its own label. We report confusion matrices, ROC and precision-recall curves, learning curves, calibration, per-fold stability and attribution for every configuration, and a five-objective assignment study in which the proposed operators beat their own base algorithm but lose to an off-the-shelf many-objective control.

Keywords: predictive process monitoring; data leakage; decision-time admissibility; survival analysis; multi-criteria decision making; service level agreements

1 Introduction

An incident arrives at a service desk. Someone must decide, within seconds, which team should handle it. That decision is expensive to reverse: in the log studied here, incidents reassigned at least once take 8.95 times longer to resolve at the median, 14.53 hours against 1.62, and 41.1 % of incidents are reassigned at least once. Getting the first assignment right is therefore worth roughly a ninefold reduction in median resolution time, and the existing process fails to do so on two cases in five.

Predictive process monitoring offers to inform that decision and reports encouraging numbers for doing so [Teinemaa et al., 2019, Ceravolo et al., 2024]. The difficulty is structural. A model trained on completed cases has access to the whole case. Attributes such as the number of reassignments, the count of activity events, or the number of distinct groups that touched an incident are all recorded in the log and none is knowable at the moment of assignment. Several are near-deterministic proxies for handle time, which is exactly the quantity the service level label thresholds. A model built from them will score well and cannot be deployed, because the information it depends on does not exist when the decision it informs is taken.

Kaufman et al. [2012] formalised this as a violation of legitimacy: a feature is admissible only if it was observable before the target was determined. Kapoor and Narayanan [2023] document the consequences across 17 scientific fields and 294 papers. The pattern recurs wherever operational logs are modelled, in connectome-based neuroimaging [Rosenblatt et al., 2024], in clinical prediction [Davis et al., 2024, Ramadan et al., 2025], in software build prediction [Mishra et al., 2026], in longitudinal education modelling [Fauszt, 2026], and in financial text [Xue et al., 2026].

Process mining has begun to address it. Weytjens and Weerdt [2022] show that constructing benchmark splits without accounting for case overlap leaks information across the boundary. Abb et al. [2024] measure example leakage above 80 % across standard BPIC logs and find that a naive baseline matches a state-of-the-art deep model once that leakage is present, a result Weytjens and Weber [2026] extend to transformer and large-language-model baselines.

What is missing is a measurement, on one log under one protocol, of both what the discipline costs and what it buys, reported together. A cost figure alone invites the conclusion that admissibility is not worth paying for. A benefit figure alone is unfalsifiable. This paper reports both.

1.1 Research questions

RQ1 How much discrimination is lost when every attribute must be observable at the moment of assignment and every group-level encoding must be estimated from past cases only?

RQ2 Under that constraint, does adding attributes beyond the small sets used in prior work still improve discrimination, and by how much?

RQ3 Does choosing an admissible modelling target, rather than the convenient one, cost accuracy or gain it?

RQ4 Does a priority-aware adaptive multi-objective assignment policy improve on standard NSGA-II under the same admissibility constraint?

1.2 Contributions

1. A decision-time admissibility protocol stated as two rules and implemented as executable constraints rather than as documentation, with its cost measured at 0.0955 AUC on BPI Challenge 2014, which is 25.0 % of all above-chance signal (RQ1).
2. An admissible feature ladder rising from 0.5076 at three attributes to 0.7859 at seventeen, a gain of 0.2243 AUC obtained without any inadmissible attribute, reproduced across two split ratios to within 0.003 (RQ2).
3. Two decision-aware scoring formulations. An assignment score aggregating seven criteria by a weighted power mean whose exponent parameterises compensation between criteria, and a service level risk score obtained by modelling resolution as a discrete-time hazard and deriving breach as a survival ratio. The second reaches 0.8017 AUC at intake against 0.7399 for a direct classifier, while being structurally unable to observe its own label (RQ3).
4. A multi-objective assignment study in which the proposed operator set improves on its own base algorithm on four of five metrics but is outperformed by an off-the-shelf many-objective control, with an ablation attributing essentially the whole gain to one of its three

components. Two reproducibility defects were found in our own harness and corrected before any number in this paper was produced. The uncorrected harness reversed this conclusion (RQ4).

2 Related work

2.1 Leakage, admissibility and target encoding

Kaufman et al. [2012] give the canonical formulation and the learn-predict separation principle. Kapoor and Narayanan [2023] provide an eight-type taxonomy and connect leakage to reproducibility failure at scale. Domain treatments converge on the same shape: an accuracy collapse once only pre-decision features are admitted, with Mishra et al. [2026] reporting a 15.07 point drop on one platform after removing leaky features and arguing that prior studies reporting 95 to 99 per cent accuracy used contaminated attributes.

Target encoding deserves separate attention because it is the second of our two rules. Pargent et al. [2022] show that regularised target encoding outperforms traditional categorical handling, and the regularisation they describe is exactly what makes the encoding safe only when it is refitted inside each resampling fold. An encoding fitted once over the full dataset before splitting contaminates every subsequent evaluation and includes each row's own label in its own predictor.

Within process mining, Weytjens and Weerdt [2022] address leakage in benchmark construction and show that a plain temporal split is insufficient because training cases overlap the test period. Abb et al. [2024] quantify example leakage on the standard BPIC logs. Our protocol extends this from the split axis to the feature axis and the time axis at once, and measures the cost rather than only prescribing the constraint.

2.2 Predictive and prescriptive process monitoring

Teinemaa et al. [2019] provide the standard outcome-prediction benchmark across 24 real-world datasets, complemented by remaining-time benchmarks [Verenich et al., 2019] and deep-learning comparisons [Rama-Maneiro et al., 2023, Kratsch et al., 2021]. We note explicitly that **BPI Challenge 2014 is not in the Teinemaa dataset list**, verified against the paper itself. There is therefore no published same-task baseline on this log, so the ranges reported in those benchmarks are used here as context and never as a head-to-head comparison. Fertig et al. [2026] revisit the field under foundation models and find that tabular gradient boosting remains competitive, which is why this study uses it.

Prescriptive process monitoring asks what to do rather than what will happen [Kubrak et al., 2022]. Recent work addresses resource constraints explicitly [Shoush and Dumas, 2025] and the organisational question of whether practitioners can act on the output [Kubrak et al., 2026]. The framing matters here because an admissibility constraint is only meaningful relative to a decision, and a decision is only meaningful if someone can take it at the time the model fires.

2.3 IT service management and this log

BPI Challenge 2014 [van Dongen, 2014] has been analysed for workload impact [Dees and van den End, 2014] and with fuzzy and heuristic mining [Andreswari et al., 2023], and the BPI challenge series has been surveyed [Lopes and Ferreira, 2019]. Adjacent operational problems include incident-management prediction [Sarnovský and Surma, 2018], ticket reassignment prediction [Schad et al., 2022], assignment-group classification [van Gassel and Janssen, 2024], service level breach prediction [Gouryraj et al., 2021, Swain and Garza, 2023], and escalation with large language models [Liu et al., 2025]. Ticket assignment has also been posed as a multi-objective problem [Arain and Mazhar, 2026], which is the closest prior framing to our Section on assignment.

2.4 Multi-objective assignment and its known limits

NSGA-II [Deb et al., 2002] remains the reference algorithm, with NSGA-III [Deb and Jain, 2014] introduced for many-objective problems where dominance-based selection degrades. That degradation is well characterised and is still an active theoretical subject [Zheng and Doerr, 2024, Doerr et al., 2026, Opris, 2025, Alghouass et al., 2025], and empirical comparisons on applied problems report mixed outcomes [dos Santos et al., 2024, Wolschick et al., 2024, Ishibuchi et al., 2025, Wang et al., 2025a, Li et al., 2025]. Our five objectives place this study in exactly that regime, so a result in which NSGA-III outperforms an NSGA-II variant is consistent with theory rather than surprising.

Wang et al. [2025b] argue that a substantial share of claimed novelty in metaheuristic research does not survive controlled comparison. That argument is directly relevant to the result we report in Section 6.7, and it is one reason we report the ablation rather than only the aggregate.

2.5 Positioning against the closest prior work

Table 1 places this study against the work nearest to it on each of its three axes: leakage measurement, service level prediction on operational logs, and multi-objective assignment.

A word on what the table deliberately does not contain. It carries a reported-performance column only where a figure is comparable to ours, and marks the rest as not comparable rather than printing a number that would invite a false ranking. Accuracy figures across different logs, different labels and different decision points are not on a common scale, and assembling them into a league table is the specific practice that produced the leakage literature this paper builds on. **BPI Challenge 2014 is not in the Teinemaa et al. [2019] dataset list**, verified against that paper itself, so there is no published same-task baseline on this log and the ranges reported in those benchmarks are used here as context and never as a head-to-head comparison.

What is missing from every row above is a measurement, on one log under one protocol, of both what the discipline costs and what it buys, reported together. That is the gap this paper fills.

2.6 Aggregation, weighting and calibration

The weighted power mean is a compensative connective in the sense of Dyckhoff and Pedrycz [1984], generalised further by Aggarwal [2015] and given an explicit andness and orness parameterisation by Dujmovic [2015]. It is the natural aggregator when the degree of permitted compensation between criteria is itself a design question rather than an assumption.

CRITIC [Diakoulaki et al., 1995] derives weights from contrast intensity and conflict. Objective weighting methods are not interchangeable [Mukhametzyanov, 2021, Ariyanti and Fu'adi, 2025], their stability under perturbation varies [Linh et al., 2025, Paradowski et al., 2025], they are susceptible to rank reversal [Li and Abbas, 2026], and sensitivity analysis over the weight vector is recommended practice [Gaona et al., 2025]. We therefore report three schemes and a rank-stability analysis rather than a single weight vector.

Because the scores feed a downstream decision, calibration matters as much as ranking [Filho et al., 2023], and the excess risk of acting on an uncalibrated classifier is quantifiable [Perez-Lebel et al., 2025]. Explanation is subject to its own limits [Bilodeau et al., 2024, Hwang et al., 2025],

Table 1: This study against the closest prior work. A dash in the performance column means the figure exists in the cited work but is not on a scale comparable with ours, because the task, the log or the decision point differs.

Study	Approach	Data	Reported	Limitation for this question
Kaufman et al. [2012]	Formalises leakage and the learn-predict separation	Several	—	States the condition; does not measure what enforcing it costs
Kapoor and Narayanan [2023]	Eight-type leakage taxonomy across 17 fields	294 papers	—	Survey of consequences, not a measurement on one log
Abb et al. [2024]	Measures example leakage in process mining	Standard BPIC logs	$> 80\,\%$ example leakage	Measures the split axis; does not price the feature axis
Mishra et al. [2026]	Removes leaky features from build prediction	CI build logs	-15.07 accuracy points	Different domain; no benefit side reported
Weytjens and Weerdt [2022]	Leak-free benchmark construction	Process logs	—	Prescribes the constraint rather than costing it
Teinemaa et al. [2019]	Outcome-prediction benchmark	24 event logs	—	BPI 2014 not among the datasets
Dees and van den End [2014]	Workload impact analysis on this log	BPI 2014	—	Descriptive; no decision-time constraint
Gouryraj et al. [2021]	Service level breach prediction	ITSM tickets	—	No admissibility constraint stated
Arain and Mazhar [2026]	Multi-objective ticket assignment	ITSM	—	No leakage control on the objective models
Deb and Jain [2014]	NSGA-III for many-objective problems	Benchmarks	—	Not applied to assignment under admissibility
This work, unconstrained	Reproduction of the default practice	BPI 2014, 31 236	0.8814 AUC	Reported only to size the correction
This work, admissible	Rules 1 and 2 enforced	BPI 2014, 31 236	**0.7859** AUC	The deployable figure
This work, chronological	Admissible, evaluated forward in time	BPI 2014, 31 236	0.6533 AUC	The figure under drift

and in process monitoring specifically the guidance is to obtain interpretable models rather than to explain opaque ones after the fact [Stevens and Smedt, 2024, Mehdiyev et al., 2025, Sahatova et al., 2026]. Whether explanations actually improve human decisions is not settled [Haag, 2026, Chae et al., 2025].

3 Data and the admissibility protocol

3.1 The log

BPI Challenge 2014 [van Dongen, 2014] is the incident-management record of Rabobank Nederland Group ICT, released for the 2014 Business Process Intelligence Challenge. It is one of the few public logs in which the assignment decision, the resource that received it and the eventual outcome are all recorded on real work rather than simulated. Three files are used: an incident header, an activity log and an interaction log. Table 2 reports the working statistics.

Table 2: Dataset statistics. Counts are produced by `evaluation/deep_metrics.py` and written to `dataset_stats.csv`; nothing in this table is typed by hand.

Property	Value
Raw incident records	31 238
Raw activity events	210 837
Raw interaction records	68 358
Incidents with a resolvable handle time and priority	31 236
Incidents in the predictive modelling matrix	31 236
Distinct first-assignment groups	117
Distinct configuration-item types	13
Distinct configuration-item subtypes	58
Distinct service components (WBS codes)	259
Admissible attributes after the protocol	17
Attributes excluded as inadmissible	9
Breach cases (positive class)	11 444
Non-breach cases (negative class)	19 792
Positive class rate	0.3664
Training incidents, random 70/30 split	21 865
Test incidents, random 70/30 split	9371
Training incidents, chronological holdout	24 988
Test incidents, chronological holdout	6248
Earliest incident open time	2012-01-10
Latest incident open time	2014-12-01
Observation window	1055 days

Four properties of the raw files silently corrupt a naive load, and they are recorded here because they cost real debugging time and because a reader attempting to reproduce this work will meet all four. The files are semicolon delimited rather than comma delimited. They are encoded latin-1 rather than UTF-8, so the group names carry mojibake under a UTF-8 read. The handle-time column uses a decimal comma, so a direct numeric cast silently produces nulls. And the incident header pads to 78 columns with empty trailing delimiters, which shifts the column index of every field a naive parser reads positionally.

3.2 Two working sets, and why they differ

The paper has two components with different data requirements, and they operate on different subsets of the same log. Reporting one figure for both would be wrong, so both are stated.

The *predictive* component uses every incident with a resolvable handle time and a recorded priority, which is 31 236 incidents across 117 distinct first-assignment groups. It places no constraint on group size, because a target encoding with a smoothing prior handles a rarely used group correctly by shrinking it towards the population rate.

The *assignment* component of Section 6.7 simulates a decision over a candidate set, and a candidate set containing groups that have handled three incidents in three years is not a realistic decision. It therefore restricts to assignment groups with at least 50 recorded cases, which leaves 15 828 incidents, 50 groups and 47 configuration-item categories, of which the earliest 12 662 by open time form the training period on which every parameter is fitted.

3.3 The service level label

The log's own `Alert Status` field reads `closed` on 46 606 of the 46 809 raw records and carries no usable breach signal, so no contractual service level target is recoverable from the data. The label is therefore constructed, and the construction is stated precisely rather than described loosely.

Let H_i be the recorded handle time of incident i in hours and $\pi(i)$ its priority class. Write $\tilde{H}_p$ for the median handle time of priority class p. An incident breaches when

$$y_i \;=\; \mathbb{1}\big[\,H_i \;>\; \tau_{\pi(i)}\,\big], \qquad \tau_p \;=\; 2\,\tilde{H}_p. \tag{1}$$

The factor of two is a deliberate choice of a lenient target: at a factor of one the positive class would be half the data by construction and the label would carry no information beyond the median. Table 3 gives the resulting thresholds. The breach rate is notably flat across priority classes, between 33.6 % and 37.6 %, which is a direct consequence of defining the threshold within each class: the label measures whether a case was slow *for its own priority*, not whether it was slow in absolute terms.

Table 3: Per-priority volume, median handle time, derived service level threshold and breach rate. Priority 1 does not occur in the filtered log.

Priority	Incidents	Median handle (h)	Threshold τ_p (h)	Breach rate
2	521	3.68	7.36	0.3359
3	4347	1.58	3.17	0.3759
4	14 810	4.05	8.10	0.3629
5	11 558	8.37	16.74	0.3686

Where the threshold is estimated, and where that matters. For the random-split arms, $\tilde{H}_p$ is computed over the whole log. This is a property of the *label*, not of a feature: no model observes $\tilde{H}_p$, and every model is scored against the same fixed label, so the comparison between configurations is unaffected. For the chronological arm it would nevertheless be wrong, because a system deployed in January 2014 cannot know the 2014 medians. That arm therefore re-derives τ_p from training-period cases alone, and the effect is reported in Section 5.11.

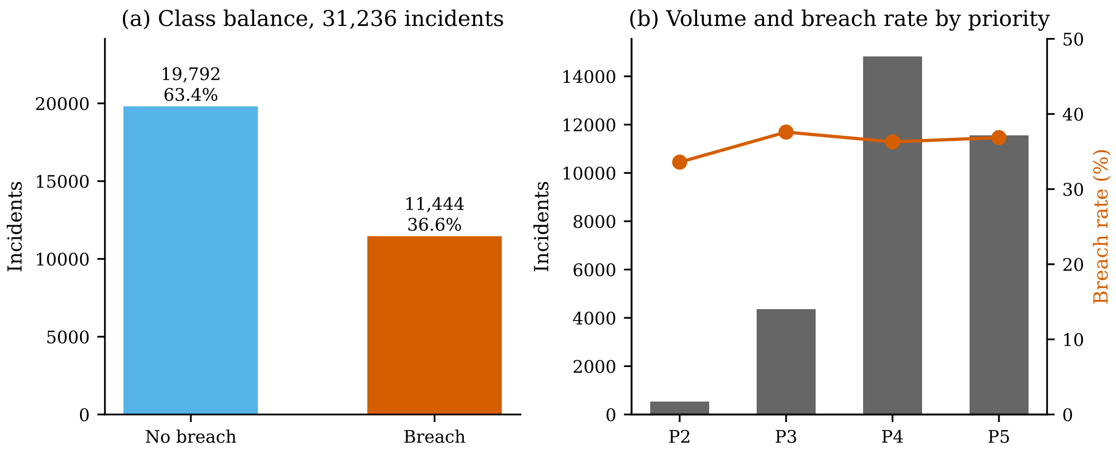

Figure 1: Class balance and priority composition. The positive class is a 36.6 % minority, which is imbalanced enough that accuracy alone would be misleading and is why precision, recall, the Matthews correlation coefficient and average precision are reported throughout.

3.4 The protocol

Let t_i^{dec} be the moment the assignment decision for case i is taken, and let $\mathcal{F}_t$ be the information available to the organisation at time t.

Rule 1, feature admissibility. An attribute $x^{(k)}$ is admissible for case i only if its value is $\mathcal{F}_{t_i^{\text{dec}}}$-measurable:

$$x_i^{(k)} \in \mathcal{F}_{t_i^{\text{dec}}} \quad \text{for every case } i. \tag{2}$$

Any attribute aggregated over the whole closed case violates this by construction, whatever its apparent predictive value. This is Kaufman et al.'s legitimacy condition [Kaufman et al., 2012] instantiated at a specific decision point rather than stated in general.

Rule 2, encoding admissibility. A group-level target encoding for group g evaluated for case i may use only cases resolved strictly before case i opened:

$$\hat{\theta}_g(i) \;=\; \frac{\sum_{j \in \mathcal{P}(i,g)} y_j \;+\; \alpha \bar{y}_{\mathcal{P}}}{|\mathcal{P}(i,g)| \;+\; \alpha}, \tag{3}$$

where the admissible past set is

$$\mathcal{P}(i,g) = \{\, j : g_j = g,\ t_j^{\text{res}} < t_i^{\text{open}} \,\}, \tag{4}$$

with smoothing prior $\alpha = 20$ and $\bar{y}_{\mathcal{P}}$ the prior breach rate over the same past set. Operationally this means the encoding is refitted inside every cross-validation fold and again on the outer training split, never once over the full data before splitting. Pargent et al. [2022] show that regularised target encoding outperforms conventional categorical handling; the regularisation is exactly what makes refitting necessary, because an encoding fitted once over everything contains each row's own label inside its own predictor.

Applying both rules to BPI 2014 admits 17 attributes and excludes nine. Table 4 lists every one with the reason for its verdict. The table is generated from `feature_justifications_leakfree.csv`, which the pipeline writes with a `Known_at_assignment_time` column, so the verdict is a machine-checked property of the code rather than a claim in prose.

Table 4: Attribute admissibility. Seventeen attributes are admitted, nine excluded. The two target encodings are admissible only under Equation 3; used the way version one of this pipeline used them, they are the single largest source of contamination.

Attribute	Admitted	Reason
Priority	yes	Impact times urgency as recorded at open. The service level clock is set per priority band.
Impact	yes	Business impact level recorded at open.
Urgency	yes	Urgency level recorded at open, independent of impact.
Open hour	yes	Hour of arrival. Proxies shift staffing, which governs pick-up speed.
Open day of week	yes	Weekday of arrival. Captures the weekly demand cycle.
Is weekend	yes	Reduced staffing materially changes achievable response time.
Is business hours	yes	Out-of-hours arrivals queue until staffed.
Open month	yes	Seasonal load and release-cycle effects.
Queue length at open	yes	Incidents already open at arrival. Computed from other cases only.
Assignment delay hours	yes	Elapsed hours from open to first assignment. Observable at the decision; not total handle time.
CI type	yes	Asset class of the affected configuration item.
CI subtype	yes	Finer-grained asset class.
Service component (WBS)	yes	Work breakdown code of the affected service.
Category	yes	Incident category assigned at intake.
Alert status	yes	Alert state on the record at open.
Group breach rate (encoded)	yes†	Smoothed historical breach rate of the first assignment group, past cases only, prior 20.
Group volume (encoded)	yes†	Case volume of the group, past cases only. Proxies capacity and experience.
Number of reassignments	**no**	Total over the whole closed case. Rises with duration, which defines the label.
Groups touched	**no**	Distinct groups over the whole closed case. Post hoc.
Total activity events	**no**	Every activity event across the closed case. Strongest duration proxy present.
Has reopen	**no**	Reopens occur late in a case. Post hoc.
Related incidents	**no**	Final tally on the closed record, not the value visible at open.
Interaction count	**no**	Aggregated across the full case from the interaction log. Post hoc.
First-call-resolution rate	**no**	Resolution outcome is known only after handling.
Group mean handle time (v1 form)	**no**	Mean of the quantity the label thresholds, over the full data before the split.
Group breach rate (v1 form)	**no**	Mean of the target over the full data including test rows and the row's own label.

†Admissible only under Equation 3. The same two quantities appear in the excluded list in the form version one of this pipeline computed them.

4 Methods

4.1 Notation

Write $i \in \mathcal{I}$ for an incident and $j \in \mathcal{G}$ for an assignment group, with $n = |\mathcal{I}|$ the number of incidents in an arrival batch and $m = |\mathcal{G}|$ the number of eligible groups. Case i carries an attribute vector $\mathbf{x}_i \in \mathbb{R}^d$, a binary service level label y_i from Equation 1, an open time t_i^{open}, a first-assignment time t_i^{dec} and a resolution time t_i^{res}. The decision variable $z_{ij} \in \{0, 1\}$ is one when incident i is assigned to group j. A hat denotes a quantity estimated from data.

4.2 Predictive layer

Four gradient-boosted and bagged tree ensembles are evaluated: XGBoost [Chen and Guestrin, 2016], LightGBM [Ke et al., 2017], CatBoost [Prokhorenkova et al., 2018] and a random forest [Breiman, 2001]. Tree ensembles rather than deep sequence models are used deliberately. Fertig et al. [2026] revisit predictive process monitoring under foundation models and find tabular gradient boosting still competitive, and Grinsztajn et al. [2022] give the general result that tree ensembles remain ahead of neural networks on tabular data of this size. Since the object of study here is the effect of a feature-admissibility constraint, the learner is a control variable and is held fixed across configurations rather than tuned per configuration.

Class imbalance is handled inside each learner rather than by resampling, so that no synthetic rows enter a leakage study: XGBoost receives `scale_pos_weight` set to the training negative-to-positive ratio, LightGBM uses `is_unbalance`, CatBoost uses balanced automatic class weights, and the random forest uses balanced class weights. Table 5 gives the full configuration.

4.3 Evaluation metrics

Write TP, FP, TN and FN for the four confusion-matrix cells at a given decision threshold. The reported quantities are

$$\text{Precision} = \frac{\text{TP}}{\text{TP} + \text{FP}}, \qquad \text{Recall} = \frac{\text{TP}}{\text{TP} + \text{FN}}, \tag{5}$$

$$\text{Specificity} = \frac{\text{TN}}{\text{TN} + \text{FP}}, \qquad F_1 = \frac{2\,\text{TP}}{2\,\text{TP} + \text{FP} + \text{FN}}, \tag{6}$$

and balanced accuracy $\text{BA} = \frac{1}{2}(\text{Recall} + \text{Specificity})$. We also report the Matthews correlation coefficient, which is the only one of these that degrades under *both* kinds of error on *both* classes and is therefore the single most informative scalar on an imbalanced problem [Chicco and Jurman, 2020]:

$$\text{MCC} = \frac{\text{TP} \cdot \text{TN} - \text{FP} \cdot \text{FN}}{\sqrt{D}}, \tag{7}$$

where the normalising denominator is

$$D = (\text{TP} + \text{FP})(\text{TP} + \text{FN}) \times (\text{TN} + \text{FP})(\text{TN} + \text{FN}). \tag{8}$$

Threshold-free ranking quality is ROC-AUC, and because the positive class is a 36.6 % minority we also report average precision, the area under the precision-recall curve, whose chance level is the class prevalence rather than 0.5. Probability quality is the Brier score, $\text{BS} = N^{-1} \sum_i (\hat{p}_i - y_i)^2$, reported because the scores feed a downstream decision and a ranking that is correct but miscalibrated still produces the wrong action [Filho et al., 2023, Perez-Lebel et al., 2025].

Table 5: Experimental configuration. Identical across every configuration and every split, so that differences between configurations are attributable to the admissibility constraint and not to tuning.

Setting	Value
XGBoost	300 trees, max depth 6, learning rate 0.05, subsample 0.8, column subsample 0.8, `scale_pos_weight` $= n_-/n_+$, logloss objective
LightGBM	300 trees, max depth 6, learning rate 0.05, subsample 0.8, column subsample 0.8, `is_unbalance`
CatBoost	300 iterations, depth 6, learning rate 0.05, balanced automatic class weights
Random forest	300 trees, max depth 12, balanced class weights
Primary split	Stratified random, 70/30, seed 42
Replication split	Stratified random, 80/20, same seed
Causal split	Chronological, earliest 80 % by open time trains, latest 20 % evaluates
Cross-validation	Stratified 5-fold on the training split, target encoding refitted inside every fold
Target-encoding prior	$\alpha = 20$
Decision threshold	0.5 by default; swept in Section 5.9; tuned on a held-out slice of the training period for the chronological arm only
Bootstrap resamples	2000, percentile intervals
Seed	42 throughout; two invocations produce bitwise-identical metrics
Hardware	28-thread x86-64 CPU, no GPU used at any point
Software	Python 3.11, scikit-learn, XGBoost, LightGBM, CatBoost, SHAP, pymoo 0.6.2

Two conventions are used throughout. First, *above-chance signal* means AUC − 0.5, so a drop from 0.8814 to 0.7859 is a loss of 0.0955 out of 0.3814, which is 25.0 %. Second, every interval reported is a percentile bootstrap interval over 2000 resamples of the test set, paired across configurations where a difference is quoted.

4.4 F1: the adaptive assignment score

The assignment score $A(i, j)$ maps an incident-group pair to $[0, 1]$ by aggregating seven criteria, each of which is normalised to $[0, 1]$ and each of which is either a model output or a fitted function of observable quantities. Nothing in it is a hand-chosen constant.

s_1**, outcome propensity.** The isotonic-calibrated probability that the case is resolved first time right by group j, taken from the predictive layer of Section 4.2.

s_2**, service level survival.** $s_2(i, j) = 1 - \hat{P}(\text{breach} \mid i, j)$, taken from F2 in Section 4.5.

s_3**, expertise match.** Let $\mathbf{r}_i$ be the one-hot requirement vector of the incident's configuration-item category and $\mathbf{e}_j$ the group's historical specialisation vector, the distribution of categories it has handled during the training period. Expertise is the cosine similarity

$$s_3(i,j) \;=\; \frac{\mathbf{e}_j \cdot \mathbf{r}_i}{\|\mathbf{e}_j\|_2\,\|\mathbf{r}_i\|_2} \;\in\; [0,1]. \tag{9}$$

Cosine rather than a raw count because it is invariant to group volume: a large group that has handled every category is not thereby a specialist in any of them.

s_4**, workload headroom.** Over the post-assignment load vector $\boldsymbol{\ell} = (\ell_1, \ldots, \ell_m)$, headroom is Jain's fairness index [Jain et al., 1984]

$$s_4 \;=\; \frac{\left(\sum_{k=1}^{m} \ell_k\right)^2}{m \sum_{k=1}^{m} \ell_k^2} \;\in\; \left(\tfrac{1}{m},\, 1\right]. \tag{10}$$

It is bounded, scale-free and equals one exactly at perfect balance, which the standard deviation is not and does not.

s_5**, freshness.** The probability of a favourable outcome decays with the delay before first touch. Fitting an exponential $\exp(-\lambda t)$ by maximum likelihood on the training period gives $\hat{\lambda} = 0.0191\ \mathrm{h}^{-1}$, 95 % interval $[0.0161, 0.0226]$, on $n = 7041$ cases. That is a half-life of 36.3 hours.

s_6**, urgency.** A logistic ramp in time remaining against the service level target, $s_6 = \big(1 + \exp(k(\tau - \tau_0))\big)^{-1}$, with $\hat{k} = 0.0248$ and $\hat{\tau}_0 = -5.63$ hours fitted on the same training period.

s_7**, importance.** A weighted geometric blend of normalised priority, impact and urgency,

$$s_7(i) \;=\; \prod_{c \in \{\mathrm{pri},\mathrm{imp},\mathrm{urg}\}} \tilde{c}_i^{\,\omega_c}, \qquad \sum_c \omega_c = 1, \tag{11}$$

geometric rather than arithmetic so that a low value on one axis cannot be bought back completely by a high value on another.

4.4.1 Aggregation by weighted power mean

The seven criteria are combined by the weighted power mean of order p [Dyckhoff and Pedrycz, 1984, Dujmovic, 2015]:

$$A(i,j) \;=\; \Big(\sum_{k=1}^{7} w_k\, s_k(i,j)^{\,p}\Big)^{1/p}, \qquad \sum_k w_k = 1. \tag{12}$$

The exponent is the point of the formulation, because it makes the degree of permitted compensation between criteria an explicit parameter rather than a hidden assumption. At $p = 1$ the mean is arithmetic and fully compensatory, so a strong criterion offsets a weak one; as $p \to 0$ it converges to the weighted geometric mean; at $p = -1$ it is harmonic; and as $p \to -\infty$ it converges to the minimum and becomes fully non-compensatory. Reporting a sensitivity curve over p therefore turns the qualitative claim that one bad criterion should or should not sink an assignment into a measured one.

Degeneracy, and the floor it forces. For $p \leq 0$ the mean is undefined or identically zero whenever any criterion is exactly zero. Cosine expertise *is* exactly zero for a group with no training history in the incident's category, and this occurs in 50 102 of 158 300 evaluation cells, which is 31.65 %. An explicit floor of $\varepsilon = 10^{-3}$ is therefore applied before any $p \leq 0$ aggregation. It is reported as a parameter of the method rather than buried as an implementation detail, because it changes the value of the score in a third of all cells.

4.4.2 Weighting

Three weighting schemes are computed and all three are reported. CRITIC [Diakoulaki et al., 1995] derives weight from contrast intensity times conflict. With the decision matrix min-max normalised to $\mathbf{Z}$, standard deviation σ_k per column and Pearson correlation $\rho_{k\ell}$ between columns,

$$w_k^{\text{CRITIC}} \;\propto\; \sigma_k \sum_{\ell=1}^{7} \bigl(1 - \rho_{k\ell}\bigr). \tag{13}$$

Entropy weighting takes $P_{ik} = Z_{ik} / \sum_i Z_{ik}$, computes $E_k = -(\ln N)^{-1} \sum_i P_{ik} \ln P_{ik}$ and sets $w_k^{\text{ent}} \propto 1 - E_k$. Uniform weighting sets $w_k = 1/7$.

CRITIC alone would be unsafe here, and the reason is structural rather than a matter of taste. Two of the seven criteria are model outputs, so their spread reflects the calibration of a classifier rather than the importance of a criterion, and isotonic calibration compresses variance. Reporting CRITIC alone would launder a modelling artifact into a data-driven weight. The question is therefore settled empirically, by measuring whether the induced assignment ordering depends on the scheme at all (Section 6.3).

4.5 F2: service level risk as a derived quantity

Modelling breach directly as the survival event is degenerate on this log, and this was established empirically before the design was changed rather than assumed. Because the label is defined by handle time exceeding a per-priority threshold, a case that has been open longer than its own threshold has already breached: the measured breach rate past threshold is exactly 1.0000. A hazard model given elapsed time and priority therefore reads its label rather than predicting it. An early version of this model scored 0.9418 AUC while learning nothing, and that number is reported here as a worked example of the failure mode the paper is about.

The admissible object is the hazard of *resolution*, which elapsed time does not determine. Each case contributes one row per elapsed-time bin it survives into, over the ten bins with edges at $0, 1, 2, 4, 8, 16, 24, 48, 96, 168$ hours and an open final bin. The expansion turns 15 828 cases into 69 867 person-period rows, a factor of 4.41, with a per-bin resolution rate of 0.2260. A gradient-boosted classifier estimates the discrete hazard

$$h(t \mid \mathbf{x}) \;=\; P\bigl(t^{\text{res}} \in b_t \bigm| t^{\text{res}} \geq b_t,\, \mathbf{x}\bigr) \tag{14}$$

directly, which is the discrete-time survival formulation of Suresh et al. [2022]. Survival follows by the product rule and breach risk is the conditional survival ratio:

$$S(t \mid \mathbf{x}) = \prod_{u \leq t} \bigl(1 - h(u \mid \mathbf{x})\bigr), \tag{15}$$

$$P\bigl(\text{breach} \mid \text{alive at } t, \mathbf{x}\bigr) = \frac{S(\tau \mid \mathbf{x})}{S(t \mid \mathbf{x})}, \tag{16}$$

where τ is the case's own service level threshold from Equation 1. Monotonicity of S is asserted as a test rather than assumed, and the test passes.

The covariates are priority, impact, urgency, assignment delay, category, the bin index and the bin start time, all of which satisfy Equation 2. The escalation cut point is tuned on training-period data alone and then frozen at 0.5305; tuning it on the evaluation period would reproduce, on the time axis, exactly the leak this paper measures on the feature axis.

4.6 Multi-objective assignment

A batch of n incidents must be assigned across m eligible groups. The formulation is

$$\max f_1(\mathbf{z}) = \tfrac{1}{n}\textstyle\sum_{i,j} z_{ij}\,\hat{p}_{ij}^{\text{ftr}} \tag{17}$$

$$\min f_2(\mathbf{z}) = \tfrac{1}{n}\textstyle\sum_{i,j} z_{ij}\,\hat{p}_{ij}^{\text{breach}} \tag{18}$$

$$\min f_3(\mathbf{z}) = \operatorname{sd}\big(\ell_1, \ldots, \ell_m\big), \quad \ell_j = \textstyle\sum_i z_{ij} \tag{19}$$

$$\max f_4(\mathbf{z}) = \tfrac{1}{n}\textstyle\sum_{i,j} z_{ij}\,s_3(i,j) \tag{20}$$

$$\min f_5(\mathbf{z}) = \tfrac{1}{n}\textstyle\sum_{i,j} z_{ij}\,\hat{d}_{ij} \tag{21}$$

where f_1 is expected first-time-right resolution, f_2 expected service level breach, f_3 workload imbalance, f_4 expertise match and f_5 expected delay before first touch. The formulation is subject to

$$\text{C1:}\ \textstyle\sum_j z_{ij} = 1 \qquad \forall i \tag{22}$$

$$\text{C2:}\ \textstyle\sum_i z_{ij} \le \kappa_j \qquad \forall j \tag{23}$$

$$\text{C3:}\ z_{ij} \le a_j \qquad \forall i, j \tag{24}$$

$$\text{C4:}\ z_{ij}\,s_3(i,j) \ge \eta\, z_{ij} \qquad \forall i, j \tag{25}$$

with $z_{ij} \in \{0, 1\}$. C1 assigns each incident exactly once, C2 respects the capacity κ_j of group j, C3 restricts assignment to available groups through the indicator a_j, and C4 imposes a minimum expertise eligibility η. Constraint C2 is enforced by a bounded repair operator rather than by penalty, so that no infeasible solution can enter the population; the operator is capped at six passes and its effect is verified by counting violations, which are zero across all 1225 runs.

Workload imbalance is kept as the standard deviation in f_3. Jain's index of Equation 10 is reported *alongside* it for every policy and never substituted for it, because changing an objective after observing that a method loses on it would be fitting the metric to the result.

4.6.1 The proposed operators

Three modifications to NSGA-II [Deb et al., 2002] are proposed and, critically, are ablated individually rather than reported as a bundle.

Priority-aware initialisation. A fraction of the initial population is constructed greedily in descending order of incident priority rather than sampled uniformly, so that the search starts from solutions in which urgent cases already sit with capable groups.

Adaptive crossover. The crossover probability rises when population diversity, measured as the mean pairwise assignment Hamming distance, falls below a threshold.

Service-level-aware mutation. Mutation is biased towards reassigning the incidents with the highest current breach probability.

Hypervolume is computed against a reference point fixed as the nadir across all variants and all runs within a batch, in a first pass completed before any hypervolume is taken. This matters: the alternative, taking the reference point from whichever run finished first, makes every hypervolume value depend on execution order, and correcting it reversed the sign of this study's headline comparison (Section 7.9).

Algorithm 1 Priority-aware adaptive NSGA-II (PAA-NSGA-II). Differences from standard NSGA-II are lines 1, 6 and 7.

1: $P_0 \leftarrow \textsc{PriorityAwareInit}(N, \rho)$ ▷ ρ of N built greedily by descending priority
2: $P_0 \leftarrow \textsc{CapacityRepair}(P_0)$
3: **for** $g = 1$ **to** G **do**
4: $\quad \delta \leftarrow \textsc{Diversity}(P_{g-1})$ ▷ mean pairwise Hamming distance
5: $\quad Q \leftarrow \textsc{TournamentSelect}(P_{g-1})$
6: $\quad Q \leftarrow \textsc{Crossover}(Q,\ p_c(\delta))$ ▷ p_c raised when δ is low
7: $\quad Q \leftarrow \textsc{Mutate}(Q,\ p_m,\ \text{bias} \propto \hat{p}^{\text{breach}})$
8: $\quad Q \leftarrow \textsc{CapacityRepair}(Q)$
9: $\quad R \leftarrow P_{g-1} \cup Q$
10: $\quad \mathcal{F} \leftarrow \textsc{FastNonDominatedSort}(R)$ ▷ $O(MN^2)$
11: $\quad P_g \leftarrow \textsc{SelectByRankAndCrowding}(\mathcal{F}, N)$
12: **end for**
13: **return** non-dominated front of P_G

5 Results: the predictive layer

5.1 The cost of admissibility

Table 6 reports the same experiment under three configurations for each of the four learners. The unconstrained configuration is the one a practitioner reaches for by default: every attribute present in the log, group statistics computed once over the whole dataset. The admissible configuration applies Rules 1 and 2. The chronological configuration applies them and additionally moves evaluation forward in time.

Table 6: Held-out ROC-AUC under the three configurations, 70/30 split. The cost column is the loss from enforcing admissibility, with a 95 % paired bootstrap interval over 2000 resamples. The final column expresses that loss as a share of all signal above chance, AUC − 0.5.

Learner	Unconstrained	Admissible	Chronological	Cost (95% CI)	Share of signal
XGBoost	0.8814	**0.7859**	0.6533	0.0955 [0.0838, 0.1074]	25.0 %
LightGBM	0.8795	0.7825	0.6456	0.0970 [0.0850, 0.1087]	25.6 %
CatBoost	0.8737	0.7701	0.6387	0.1036 [0.0918, 0.1165]	27.7 %
Random forest	0.8720	0.7727	0.6310	0.0993 [0.0876, 0.1115]	26.7 %
Attributes	20	17	17		

Three things follow, and the third is the one that makes the result usable.

The loss is large. Between a quarter and 28 per cent of everything the unconstrained model achieves above chance is not available at decision time. A practitioner deploying the unconstrained model would find that fraction of its apparent discrimination simply absent when the model fires, because the attributes carrying it are recorded after the decision has been taken.

The loss is not a property of one learner. All four lose between 0.0955 and 0.1036 AUC, and every 95 % interval excludes zero by a wide margin: the smallest lower bound is 0.0838, more than eight times the standard error of a single AUC estimate on this test set. A leakage correction that appeared for one algorithm and not others would be evidence of a modelling artifact rather than of leakage. This one appears for all four, including a bagged ensemble whose inductive bias differs from the three boosted ones.

The loss is not an artifact of the split. Repeating the whole protocol at 80/20 gives 0.8833 unconstrained against 0.7877 admissible, a cost of 0.0956, which agrees with the 70/30 estimate to within 0.0002 AUC.

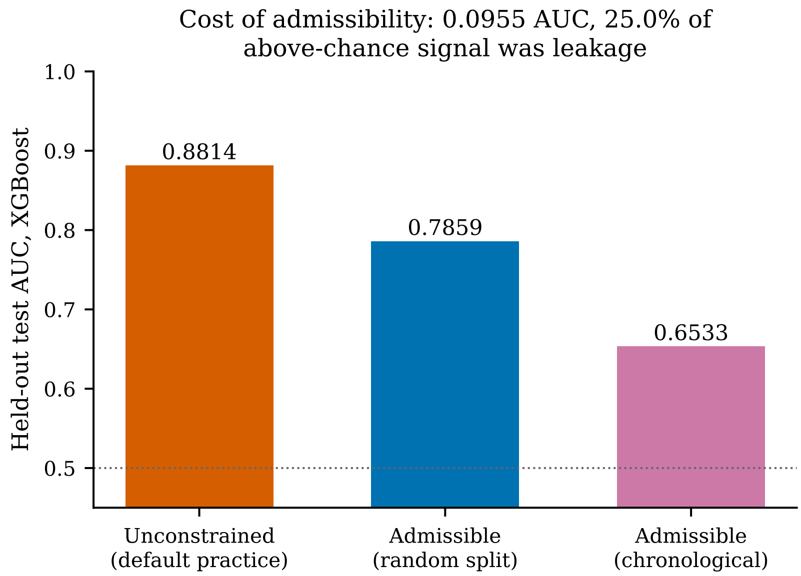

Figure 2: Where the unconstrained model's discrimination goes when each rule is applied.

5.2 The full comparison

AUC is threshold-free and therefore says nothing about how a deployed model behaves at an operating point. Table 7 reports ten metrics for every learner under every configuration at the default threshold of 0.5.

Table 7: Complete metric set, 70/30 split, decision threshold 0.5. Chance average precision is the class prevalence, 0.3664. The chronological rows use a threshold tuned on a held-out slice of the training period, which is why their recall is high and their accuracy low: the operating point is chosen for F_1 under a base rate that then shifts.

Configuration	Learner	Acc.	Prec.	Rec.	Spec.	F_1	MCC	AUC	AP	Brier
Unconstrained	XGBoost	0.7915	0.6873	0.7906	0.7920	0.7353	0.5684	0.8814	0.8242	0.1386
Unconstrained	LightGBM	0.7867	0.6804	0.7876	0.7861	0.7301	0.5594	0.8795	0.8208	0.1403
Unconstrained	CatBoost	0.7853	0.6796	0.7833	0.7865	0.7277	0.5558	0.8737	0.8138	0.1438
Unconstrained	Random forest	0.7856	0.6872	0.7614	0.7996	0.7224	0.5504	0.8720	0.8068	0.1430
Admissible	XGBoost	**0.7213**	0.6074	0.6761	0.7474	**0.6399**	**0.4152**	**0.7859**	**0.6888**	**0.1861**
Admissible	LightGBM	0.7166	0.6020	0.6679	0.7447	0.6333	0.4048	0.7825	0.6843	0.1881
Admissible	CatBoost	0.7015	0.5818	0.6589	0.7262	0.6179	0.3766	0.7701	0.6686	0.1939
Admissible	Random forest	0.7185	0.6155	0.6170	0.7772	0.6162	0.3940	0.7727	0.6672	0.1900
Chronological	XGBoost	0.4691	0.3553	0.8858	—	0.5072	—	0.6533	—	0.2037
Chronological	LightGBM	0.4854	0.3589	0.8500	—	0.5047	—	0.6456	—	0.2029
Chronological	CatBoost	0.4627	0.3448	0.8246	—	0.4863	—	0.6387	—	0.2025
Chronological	Random forest	0.4595	0.3429	0.8210	—	0.4837	—	0.6310	—	0.2035

The ordering of learners is stable across every metric and both configurations: XGBoost first, LightGBM second, then the random forest and CatBoost trading places depending on whether the metric rewards ranking or thresholded decisions. The gap between XGBoost and LightGBM is 0.0034 AUC with a paired bootstrap interval of $[-0.0099, 0.0170]$, so it is not distinguishable; the gaps to CatBoost (+0.0159, $[0.0020, 0.0297]$) and to the random forest (+0.0134, $[0.0006, 0.0269]$) both exclude zero. XGBoost is used for every subsequent single-model analysis on that basis.

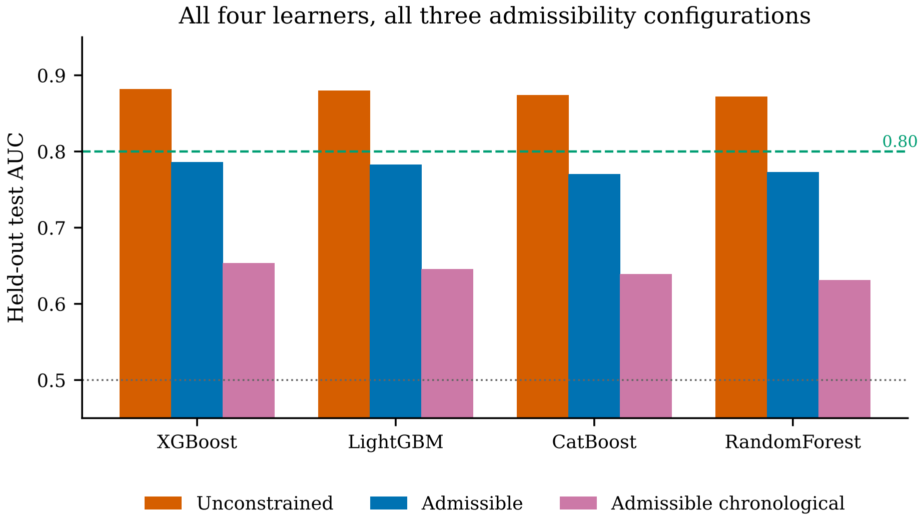

Figure 3: Learner comparison across configurations. The correction is of the same size for every learner, which is what distinguishes a leakage effect from a modelling artifact.

5.3 Where the admissible model fails

Table 8 and Figure 4 give the confusion matrices. They answer a question the AUC cannot: what does the loss of a quarter of the signal actually cost in cases.

Table 8: Confusion matrices on the 9371 held-out incidents, threshold 0.5. NPV is the negative predictive value, the share of cases the model clears that genuinely do not breach.

Configuration	Learner	TN	FP	FN	TP	Specificity	NPV	MCC
Unconstrained	XGBoost	4703	1235	719	2714	0.7920	0.8674	0.5684
Unconstrained	LightGBM	4668	1270	729	2704	0.7861	0.8649	0.5594
Unconstrained	CatBoost	4670	1268	744	2689	0.7865	0.8626	0.5558
Unconstrained	Random forest	4748	1190	819	2614	0.7996	0.8529	0.5504
Admissible	XGBoost	4438	1500	1112	2321	0.7474	0.7996	0.4152
Admissible	LightGBM	4422	1516	1140	2293	0.7447	0.7950	0.4048
Admissible	CatBoost	4312	1626	1171	2262	0.7262	0.7864	0.3766
Admissible	Random forest	4615	1323	1315	2118	0.7772	0.7782	0.3940

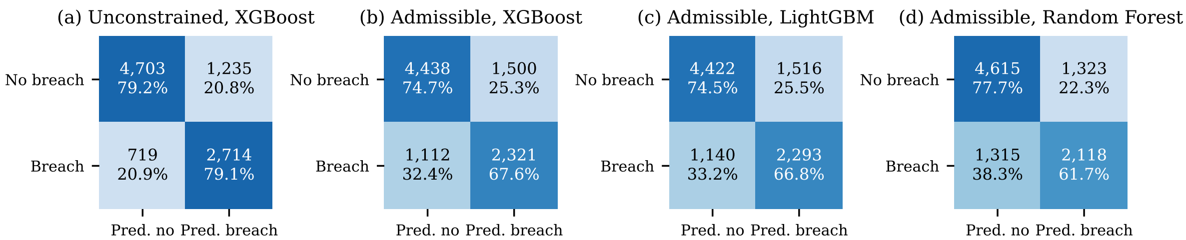

Figure 4: Confusion matrices. Cell labels give counts and row-normalised percentages. Panel (a) is the model a practitioner would build without the protocol; panel (b) is the same learner under it.

Enforcing admissibility moves 393 breaches from correctly caught to missed, taking recall from 79.1 % to 67.6 %, and adds 265 false alarms. In operational terms, an escalation queue built on the unconstrained model would appear to catch four breaches in five, and would in fact catch two in three. That difference, 393 cases in a three-month evaluation window, is the concrete form of the 0.0955 AUC.

The random forest shows the trade the other way. It has the best specificity of the four admissible models at 0.7772 and the worst recall at 0.6170: it declares breach less often, so it is right more often when it does and misses more when it does not. This is why the MCC column matters. On accuracy the random forest (0.7185) looks close to XGBoost (0.7213); on MCC, which is sensitive to both error types on both classes, the gap widens to 0.3940 against 0.4152.

5.4 Ranking behaviour across the whole operating range

Figure 5 gives ROC curves and Figure 6 precision-recall curves. The precision-recall view is the more informative of the two here, because the positive class is a 36.6 % minority and ROC curves are optimistic under imbalance.

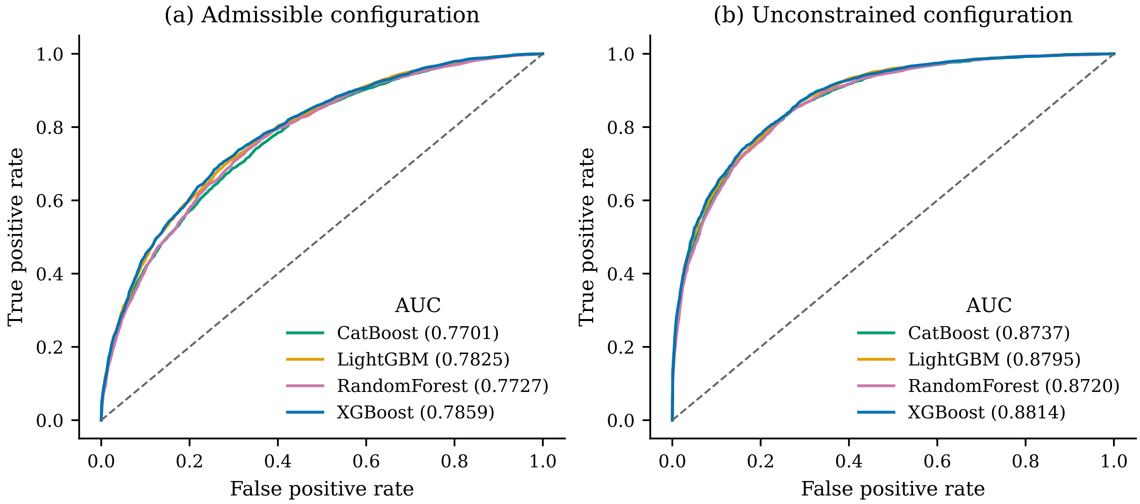

Figure 5: ROC curves for all four learners under both configurations. The four curves are almost superimposed within each panel, which is the visual form of the result that the learner choice is not what drives performance here.

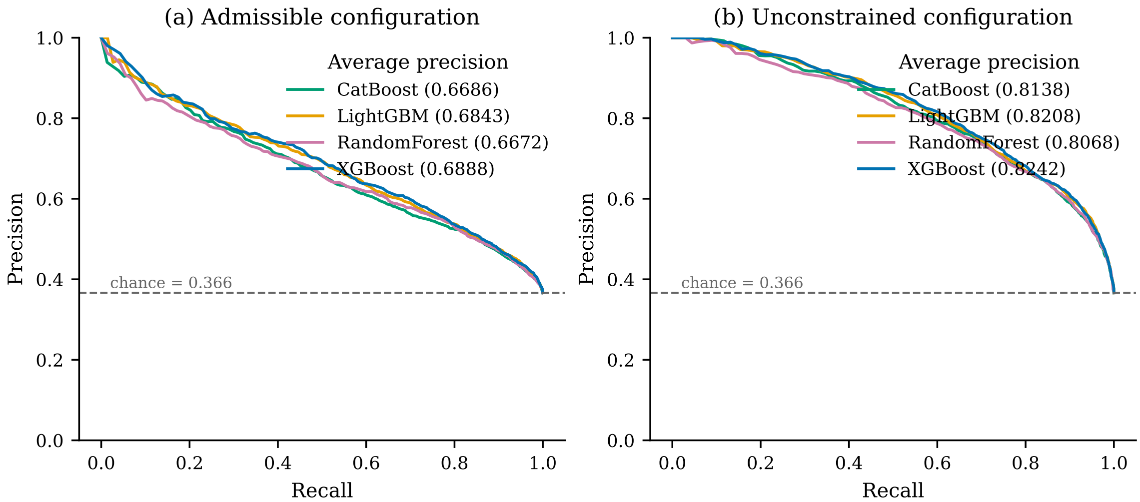

Figure 6: Precision-recall curves. The dashed line is the chance level, which under precision-recall is the class prevalence 0.3664 rather than 0.5.

Average precision falls from 0.8242 to 0.6888 for XGBoost when admissibility is enforced, a relative loss of 16.4 %, and both remain far above the chance level of 0.3664. The shape of the admissible curve is the operationally relevant part: precision holds above 0.8 only up to about 0.2 recall, then falls steadily. A service desk that wants to escalate the riskiest 10 % of arrivals can do so at roughly 82 % precision; one that wants to catch 90 % of breaches must accept precision below um0.47 and must look at seven incidents in ten.

5.5 What admissibility buys

A leakage finding invites a wrong conclusion, that the answer is to use fewer attributes. Table 9 shows it is not.

Table 9: The admissible feature ladder, XGBoost, 70/30. Every rung uses only attributes that satisfy Equation 2; no inadmissible attribute contributes to any row. This doubles as an incremental ablation of the feature groups.

Rung	Added information	k	CV-AUC	Test AUC	F_1	Precision	Recall
1	Priority, impact, urgency	3	0.5111	0.5076	0.4376	0.3725	0.5304
2	+ arrival-time context	8	0.5576	0.5616	0.4440	0.4098	0.4844
3	+ queue state and delay	10	0.6516	0.6574	0.5015	0.4814	0.5234
4	+ configuration item and category	15	0.7764	0.7767	0.6327	0.5859	0.6877
5	+ admissible group encodings	17	0.7850	**0.7859**	0.6399	0.6074	0.6761

Discrimination rises monotonically from 0.5076, which is chance, to 0.7859. The gain from eight to seventeen attributes is 0.2243 AUC, reproduced to within 0.003 at 80/20. Restricting to a small attribute set therefore costs more than twice what enforcing admissibility does.

The ladder also localises where the signal lives. The single largest jump is rung 3 to rung 4, +0.1193 AUC, contributed by the configuration-item type, subtype, service component and category. What the incident is *about* matters more than when it arrived or how busy the desk was. The admissible group encodings add 0.0092 on top, which is real but an order of magnitude smaller than the 0.0955 their inadmissible forms were worth in the unconstrained configuration. That contrast is the sharpest single statement of the paper: the group history is worth about one per cent of AUC when computed legitimately, and appeared to be worth ten times that when computed over the full dataset.

5.6 Overfitting, underfitting and how much data the problem needs

Table 10 and Figure 8 report the learning curve: training and held-out AUC for the admissible XGBoost model as the training set grows from 1093 to 21 865 incidents.

Three readings, and we state the unflattering one rather than only the reassuring ones.

The model is not underfitting. Training AUC is 0.9977 at 1093 incidents, so a 300-tree depth-6 ensemble has ample capacity to memorise this feature space; the constraint is information, not model class.

The model does overfit, and measurably. The gap at full training size is 0.0804. That is not negligible, and it is the honest answer to the question of whether 0.7859 is the model's true ability: it is the held-out estimate, and the training estimate is 0.0804 higher because 300 trees at depth 6 fit noise the test set does not contain.

The overfitting is shrinking with data, not with regularisation. The gap falls monotonically from 0.3254 to 0.0804 across a twentyfold increase in training data, and its rate of decline has not flattened: the last three points give 0.1022, 0.0896 and 0.0804. Held-out AUC over the same span rises from 0.7785 to 0.7844 and is still climbing. The practical implication is that

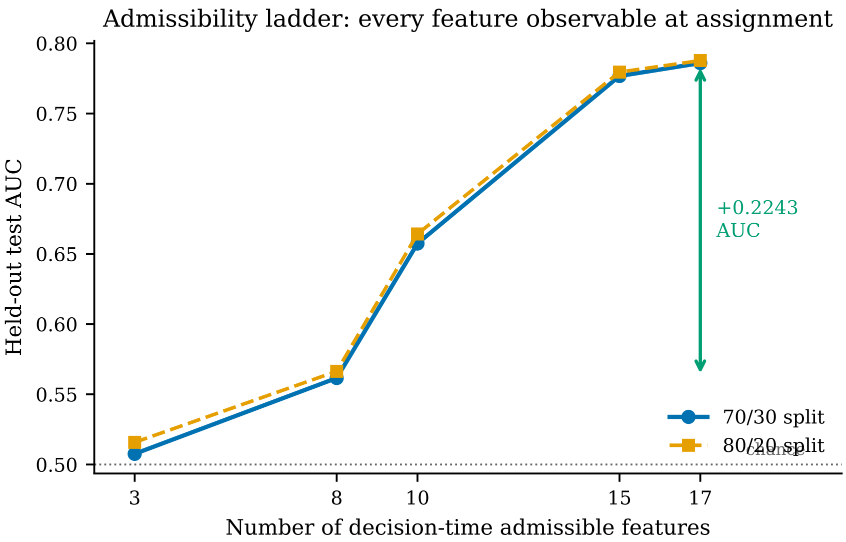

Figure 7: The admissible feature ladder under both split ratios. The two curves agree to within 0.003 AUC at every rung.

Table 10: Learning curve, admissible XGBoost. The gap is training AUC minus held-out AUC. The final row is the model reported everywhere else in this paper.

Fraction of training set	Incidents	Training AUC	Held-out AUC	Gap
0.05	1093	0.9977	0.6723	0.3254
0.10	2186	0.9868	0.7075	0.2793
0.20	4373	0.9535	0.7394	0.2141
0.30	6559	0.9282	0.7580	0.1702
0.40	8746	0.9120	0.7644	0.1476
0.50	10932	0.9042	0.7723	0.1318
0.60	13119	0.8903	0.7773	0.1131
0.70	15305	0.8807	0.7785	0.1022
0.85	18585	0.8718	0.7822	0.0896
1.00	21865	0.8647	0.7844	0.0804

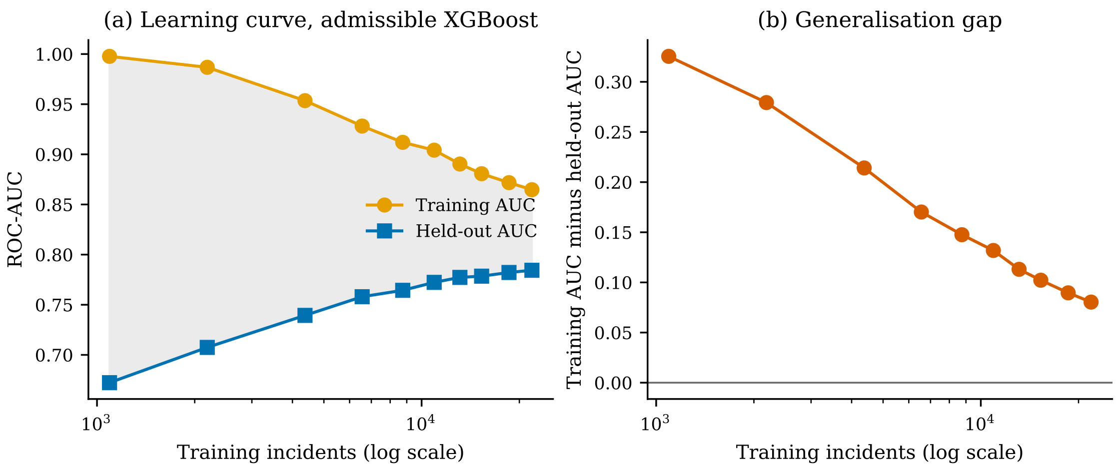

Figure 8: Learning curve and generalisation gap. The two curves converge towards each other throughout and have not met, which is the signature of a model that is still data-limited rather than capacity-limited.

more incidents from this desk would still help, and that heavier regularisation is the wrong first response, since it would move both curves down rather than move them together.

5.7 Cross-validation stability

Table 11 gives the per-fold cross-validation AUC on the training split, with the target encoding refitted inside each fold under Equation 3.

Table 11: Five-fold stratified cross-validation AUC on the admissible configuration. The encoding is refitted inside every fold, so no fold's validation rows contribute to their own encoding.

Learner	Fold 1	Fold 2	Fold 3	Fold 4	Fold 5	Mean	SD
XGBoost	0.7840	0.7869	0.7938	0.7844	0.7759	**0.7850**	0.0064
LightGBM	0.7785	0.7859	0.7890	0.7840	0.7741	0.7823	0.0060
Random forest	0.7694	0.7785	0.7788	0.7768	0.7603	0.7728	0.0079
CatBoost	0.7672	0.7723	0.7781	0.7713	0.7647	0.7707	0.0051

The cross-validation mean of 0.7850 and the held-out value of 0.7859 agree to 0.0009, so the held-out estimate is not a lucky split. Fold-to-fold standard deviation is 0.0064, an order of magnitude smaller than the 0.0955 cost of admissibility, which is what licenses treating that cost as a real effect rather than as sampling noise.

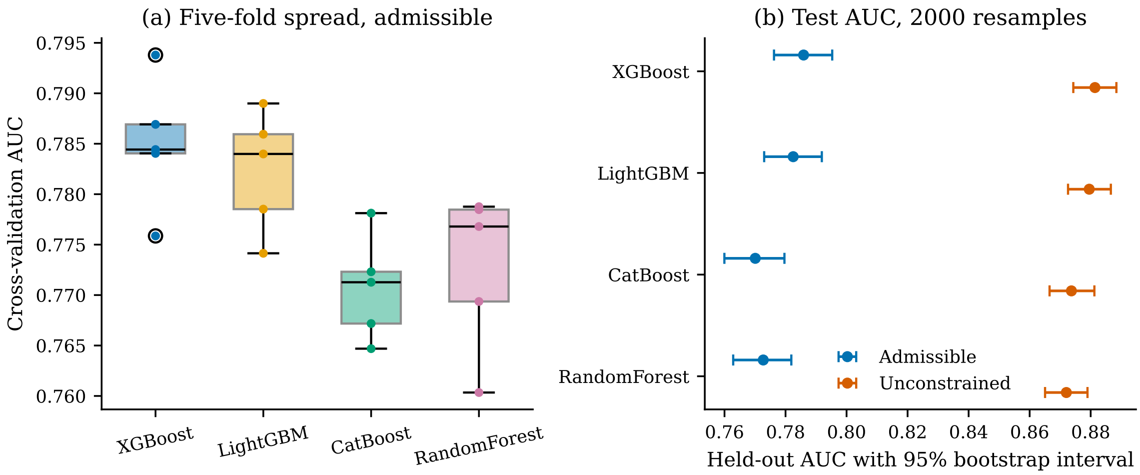

Figure 9: Fold spread and bootstrap intervals. Panel (b) shows why the two configurations are distinguishable and the four learners within a configuration largely are not: the between-configuration separation is roughly ten times the width of any single interval.

A limit of the fold-level test, stated rather than hidden. A paired Wilcoxon signed-rank test over five folds has a smallest attainable two-sided p-value of 0.0625, because there are only $2^5 = 32$ sign assignments. Every fold-level comparison here returns exactly that value, which means XGBoost beat each competitor on all five folds and that the test cannot reject at $\alpha = 0.05$ no matter how large the effect. This is a property of the test at $n = 5$, not evidence of a small effect, and it is why the model comparisons in Section 5.2 are made with 2000-resample bootstrap intervals on the test set instead.

5.8 Does tuning close the gap to 0.80?

Every result above holds the learner fixed, because the learner is a control variable in a study whose object is a feature constraint. That leaves an obvious question unanswered: is 0.7859 the ceiling of the admissible feature set, or merely the ceiling of an untuned configuration? Table 12 answers it with a randomised hyperparameter search and four ensembles under the same 70/30 protocol, with every gain tested by paired bootstrap on the same test set.

Table 12: Tuning and ensembling on the admissible feature set, 70/30. The final two columns give the paired bootstrap difference against the same model untuned, and whether its 95 % interval excludes zero. Nothing here changes the comparisons in Section 5.1, which hold the learner fixed by design.

Model	AUC	Acc.	Prec.	Rec.	F_1	Brier	Δ AUC	Sig.
XGBoost, untuned	0.7859	0.7213	0.6074	0.6761	0.6399	0.1861	—	—
LightGBM, untuned	0.7825	0.7166	0.6020	0.6679	0.6333	0.1881	—	—
CatBoost, untuned	0.7701	0.7015	0.5818	0.6589	0.6179	0.1939	—	—
Random forest, untuned	0.7727	0.7185	0.6155	0.6170	0.6162	0.1900	—	—
XGBoost, tuned	**0.7907**	0.7291	0.6183	0.6805	**0.6479**	0.1830	+0.0048	yes
LightGBM, tuned	0.7865	0.7232	0.6089	0.6834	0.6440	0.1858	+0.0005	no
Soft vote, 3 models	0.7876	0.7213	0.6076	0.6752	0.6396	0.1856	+0.0016	yes
Soft vote, 4 models	0.7863	0.7208	0.6093	0.6633	0.6351	0.1857	+0.0004	no
Stacking, logistic meta	0.7904	**0.7367**	**0.6679**	0.5596	0.6090	**0.1759**	+0.0044	yes

Significant means the 95 % paired bootstrap interval on the difference against the same model untuned excludes zero.

The answer is no, and the size of the failure is the useful part. The best tuned single model reaches 0.7907 and the best ensemble 0.7904. Both gains are real, with intervals $[0.0027, 0.0068]$ and $[0.0027, 0.0061]$ that exclude zero, and both are worth about 0.005 AUC. That is one twentieth of the 0.0955 the admissibility constraint costs. Tuning cannot buy back what the constraint removes, and it was never going to: the learning curve in Section 5.6 already showed the ceiling is information rather than model capacity, and this is the same conclusion reached from the other direction.

Two learners gain nothing. LightGBM tuned improves by 0.0005 and the four-model soft vote by 0.0004, neither distinguishable from zero. An ensemble of models that all see the same 17 attributes and all fit them with axis-aligned splits has little disagreement left to average away, which is the ordinary reason ensembling stops helping.

The stacking row is the one to read carefully, because it is the row a reader chasing a single headline number would quote. It has the best accuracy at 0.7367, the best precision at 0.6679 and the best Brier score at 0.1759, and the worst recall of any admissible model at 0.5596. It buys those three by predicting the majority class more often. On a service desk that is the wrong trade: the cost of a missed breach is an escalation that did not happen, and the cost of a false alarm is a few minutes of attention. Accuracy is the metric that rewards this behaviour and it is exactly why Table 7 does not lead with it.

We keep 0.7859 as the figure quoted throughout the rest of the paper. It is the untuned model under the same fixed configuration used for every other arm, so it is the number that makes the comparison between configurations a comparison of the constraint rather than of tuning effort. 0.7907 is what the admissible feature set yields when the learner is allowed to be optimised, and both are reported so neither has to be inferred.

5.9 The operating point

The default threshold of 0.5 is a convention, not a decision. Table 13 sweeps it.

Table 13: Threshold sweep, admissible XGBoost. Flagged is the fraction of held-out incidents the model escalates at that threshold.

Threshold	Precision	Recall	F_1	Accuracy	Flagged
0.20	0.4203	0.9729	0.5870	0.4985	0.8480
0.30	0.4691	0.9053	0.6180	0.5900	0.7070
0.40	0.5378	0.7955	0.6418	0.6746	0.5419
0.45	0.5766	0.7387	**0.6477**	0.7056	0.4693
0.50	0.6074	0.6761	0.6399	0.7213	0.4077
0.60	0.6715	0.5435	0.6008	0.7354	0.2966
0.70	0.7398	0.4026	0.5214	0.7293	0.1993
0.80	0.8227	0.2284	0.3575	0.6993	0.1017
0.90	0.9197	0.0801	0.1474	0.6604	0.0319

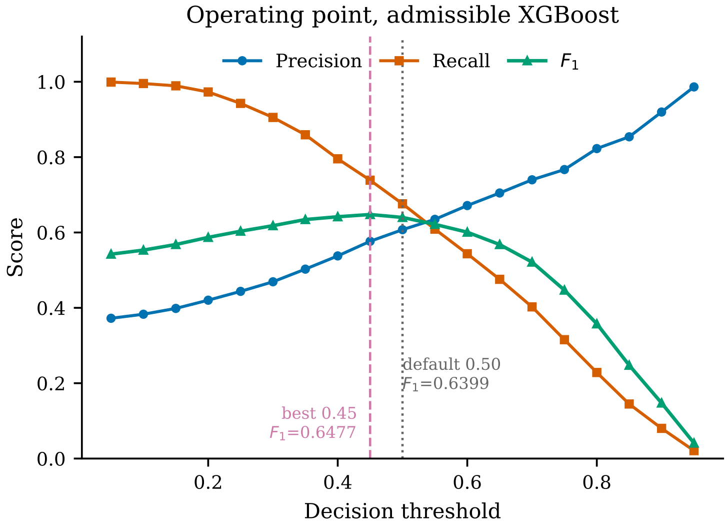

Figure 10: Precision, recall and F_1 against the decision threshold. F_1 peaks at 0.45, not at the default 0.50.

F_1 peaks at 0.6477 at a threshold of 0.45 rather than at the default, though the difference from 0.5 is 0.0078 and is not worth much on its own. The useful part of the table is the shape of the trade. At 0.90 the model escalates 3.2 % of arrivals at 92.0 % precision; at 0.30 it escalates 70.7 % of them and catches 90.5 % of breaches. A service desk with one spare senior engineer and one with a whole escalation team are choosing different points on the same curve, and neither is at 0.5. This is the argument for reporting the sweep rather than a single F_1.

5.10 Calibration

Because the scores feed a downstream decision, a correct ordering with wrong probabilities is still an operational failure [Perez-Lebel et al., 2025]. Figure 11 gives the reliability diagram before and after isotonic regression, with the calibrator fitted on out-of-fold training predictions and never on the test set.

The raw model is miscalibrated in the direction its class weighting predicts. Because `scale_pos_weight` is set to the negative-to-positive ratio, the raw scores are pushed towards the positive class and overstate breach probability in the middle of the range. Isotonic regression removes 4.9 % of the Brier score, and it is a monotone map, so AUC moves by 0.0001, which is

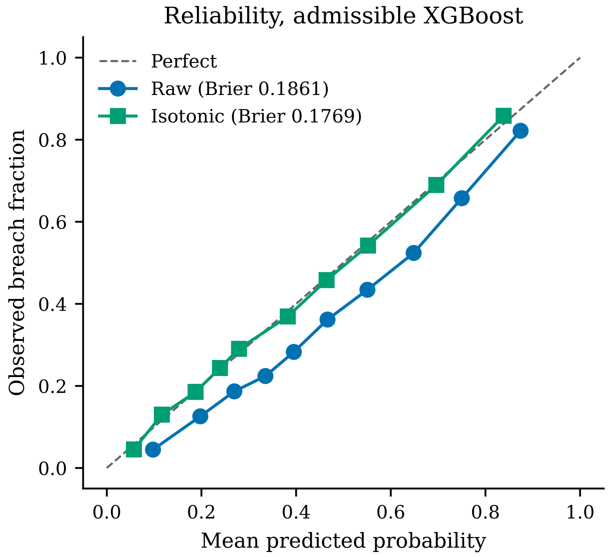

Figure 11: Reliability of the admissible XGBoost model. Isotonic calibration reduces the Brier score from 0.1861 to 0.1769 while leaving AUC unchanged at 0.7859 against 0.7858, as a monotone transform must.

the expected behaviour and a useful check that nothing else changed.

5.11 The chronological holdout

Every result above uses a random split. Event logs are temporal, and a random split lets the model train on cases that arrived after the ones it is evaluated on. The chronological arm trains on the earliest 80 % of arrivals and evaluates on the latest 20 %, re-derives the label threshold from training-period medians alone, and tunes the decision threshold on a validation slice taken from the end of the training period.

Held-out AUC falls from 0.7859 to 0.6533, a further loss of 0.1326 on top of the admissibility cost. The cause is measurable drift rather than a defect. The training period runs from 2012-01-10 to 2014-01-13 and the evaluation period from 2014-01-13 to 2014-12-01; across that boundary the median handle time falls from 4.73 to 3.37 hours and the breach rate under the causal label falls from 36.6 % to 30.8 %. The Rabobank process got faster. AUC is threshold-free, so the drop is genuine ranking degradation and not a base-rate artifact.

Table 14 separates the two failures, which are usually reported as one and are not the same thing.

At the default cut point the admissible XGBoost model recovers only 9.1 % of future breaches, and the random forest only 2.8 %: under a base rate that has fallen by six points, a threshold calibrated for the past almost never fires. That is a threshold failure, and it is fully repairable, because tuning the cut point on a validation slice from the end of the training period recovers F_1 to 0.5072 at recall 0.8858 without touching the model. The ranking failure underneath it, 0.7859 to 0.6533 AUC, is not repairable by any threshold and is the real cost of time.

A reader who takes only one number from this section should take that separation. Reporting the default-threshold F_1 of 0.1596 alone would suggest the model is useless on future data; reporting the tuned F_1 of 0.5072 alone would hide that a deployed model needs its operating point re-tuned as the process drifts. Both are true and neither is complete.

We report both arms rather than choosing one. The random split answers how well the attributes separate breach from non-breach; the chronological split answers how well a model

Table 14: Chronological holdout on 6248 future incidents. Each learner appears twice with identical predicted probabilities and therefore identical AUC; only the decision threshold differs. The default cut point is a ranking that is fine and an operating point that is wrong.

Learner	Threshold rule	Cut	Accuracy	Precision	Recall	F_1	AUC
XGBoost	default	0.50	0.7033	0.6308	0.0913	0.1596	0.6533
XGBoost	tuned on train	0.13	0.4691	0.3553	0.8858	**0.5072**	0.6533
LightGBM	default	0.50	0.7076	0.6471	0.1142	0.1941	0.6456
LightGBM	tuned on train	0.16	0.4854	0.3589	0.8500	0.5047	0.6456
CatBoost	default	0.50	0.7079	0.6564	0.1111	0.1900	0.6387
CatBoost	tuned on train	0.21	0.4627	0.3448	0.8246	0.4863	0.6387
Random forest	default	0.50	0.6977	0.7714	0.0280	0.0541	0.6310
Random forest	tuned on train	0.22	0.4595	0.3429	0.8210	0.4837	0.6310

trained today would rank tomorrow's arrivals. They are different questions and the gap between them, 0.1326 AUC, is itself the finding.

5.12 Attribution

Figure 12 gives mean absolute SHAP values [Lundberg and Lee, 2017] for the admissible model, computed on a sample of 2000 held-out incidents.

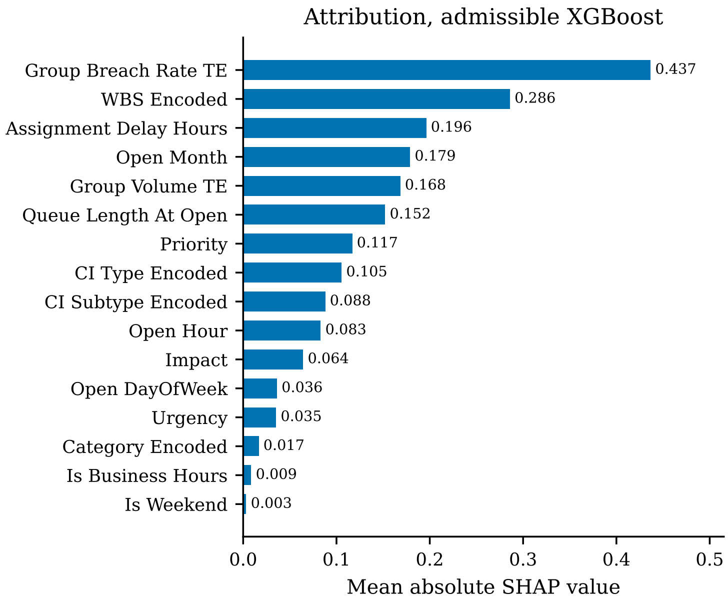

Figure 12: Mean absolute SHAP attribution, admissible XGBoost. Alert status contributes exactly zero and is retained in the table for completeness, because an attribute that survives the admissibility rules and then turns out to be uninformative is a different finding from one that was excluded.

The admissible group breach-rate encoding is the strongest single contributor at 0.4366, followed by the service component at 0.2859 and the assignment delay at 0.1962. The ordering is consistent with the feature ladder: the encodings add little *marginal* AUC at rung 5 because the configuration-item attributes at rung 4 already carry much of the same information, but conditional on being present they carry the largest per-case attribution. Marginal contribution and attribution answer different questions and it is worth saying so explicitly, because a reader who expects them to agree will read one of the two tables as an error.

Two attributions deserve comment. `Alert_Status` contributes exactly 0.0, which is consistent with the field being uniformly `closed` across almost the whole log; it passes the admissibility rules and is simply uninformative. And `Assignment_Delay` ranks third despite the corruption in its upper tail documented in the Limitations, which suggests the signal it carries is concentrated in the well-formed short-delay range.

Attribution is reported and is not validated against human decisions. SHAP has known impossibility results [Bilodeau et al., 2024] and sensitivity to feature representation [Hwang et al., 2025], and whether explanations improve human decisions is unsettled [Haag, 2026, Chae et al., 2025]. No claim is made here that these attributions improve operator performance; they are reported as a description of the model, not as an intervention.

6 Results: the scoring formulations

Section 5 measured a constraint. This section measures the two formulations built under it. The rule for both is the same and is stated once: every constant is estimated on the earliest 80 % of the timeline by incident open time, and every number reported as an evaluation is computed on the remaining 20 %, which the fitting never saw. The working set is the 15 828 incidents and 50 groups of Section 3.2, of which 12 662 form the training period.

6.1 What is fitted, and on how much

Table 15 lists every estimated constant in F1 and F2 with its interval and its sample size. Nothing in either formulation is a number chosen because it looked reasonable.

Table 15: Every fitted constant in the two formulations. Intervals are 95 %. All estimation uses training-period cases only.

Symbol	Quantity	Estimate	Interval / support
$\hat{\lambda}$	Freshness decay rate, per hour	0.0191	[0.0161, 0.0226], $n = 7041$
	implied half-life (hours)	36.3	derived from $\hat{\lambda}$
$\hat{k}$	Urgency ramp steepness	0.0248	$n = 7041$
$\hat{\tau}_0$	Urgency ramp midpoint, hours	−5.63	$n = 7041$
ε	Floor applied before any $p \leq 0$ aggregation	10^{-3}	fixed, reported as a parameter
α	Target-encoding smoothing prior	20	fixed
$\hat{\theta}$	F2 escalation cut point	0.5305	tuned on train, $F_1 = 0.4947$
	Cosine-expertise cells evaluated	158 300	3166×50
	of which exactly zero	50 102	31.65 %
	Delay rows excluded as corrupt	5627	cap at 168 h, see Limitations

Two of these deserve comment rather than a row.

The freshness decay was the first thing to break, and how it broke is worth recording. Fitted across the uncorrected `Assignment_Delay_Hours` column the maximum-likelihood estimate of λ is exactly zero with an undefined confidence interval, which is not a plausible decay rate but is a very clear signal. The cause is a corrupted upper tail in the recorded first-assignment timestamp: the median delay is 16.2 hours against a 90th percentile of 5684 hours, and the favourable-outcome rate *rises* again across that tail instead of falling. Capping at 168 hours excludes 5627 rows, 35.6 %, and produces $\hat{\lambda} = 0.0191\,\mathrm{h}^{-1}$ with a well-formed interval. The exclusion is reported rather than absorbed, because a third of a variable is a large thing to drop.

The urgency midpoint $\hat{\tau}_0 = -5.63$ hours is negative, which means the fitted logistic ramp is centred before the observed range begins and s_6 is therefore monotonically decreasing across the whole range rather than sigmoidal within it. The shape is real and it is reported as fitted. It says that on this log urgency has no inflection point inside the observable window: cases do not become suddenly urgent at some threshold, they become steadily less recoverable from the moment they arrive.

6.2 The criterion decision matrix

Table 16 gives the spread of each criterion over the evaluation period. It is the input to CRITIC, and it explains most of what the weighting analysis then finds.

Table 16: Criterion statistics over the evaluation-period decision matrix. Conflict is the CRITIC term $\sum_\ell(1-\rho_{k\ell})$. Exact zeros matter because they are what force the floor in Equation 12.

	Criterion	Mean	SD	Min	Exact zeros	Conflict
s_1	Outcome propensity	0.5276	0.3168	0.0009	0	5.363
s_2	Service level survival	0.6208	0.2738	0.0000	4093	5.371
s_3	Expertise match	0.3664	0.4247	0.0000	50 102	6.010
s_4	Workload headroom	0.3758	0.0000	0.3757	0	6.024
s_5	Freshness	0.9251	0.1741	0.0418	0	6.457
s_6	Urgency	0.4501	0.0922	0.3819	0	7.107
s_7	Importance	0.1898	0.1710	0.0010	0	5.691

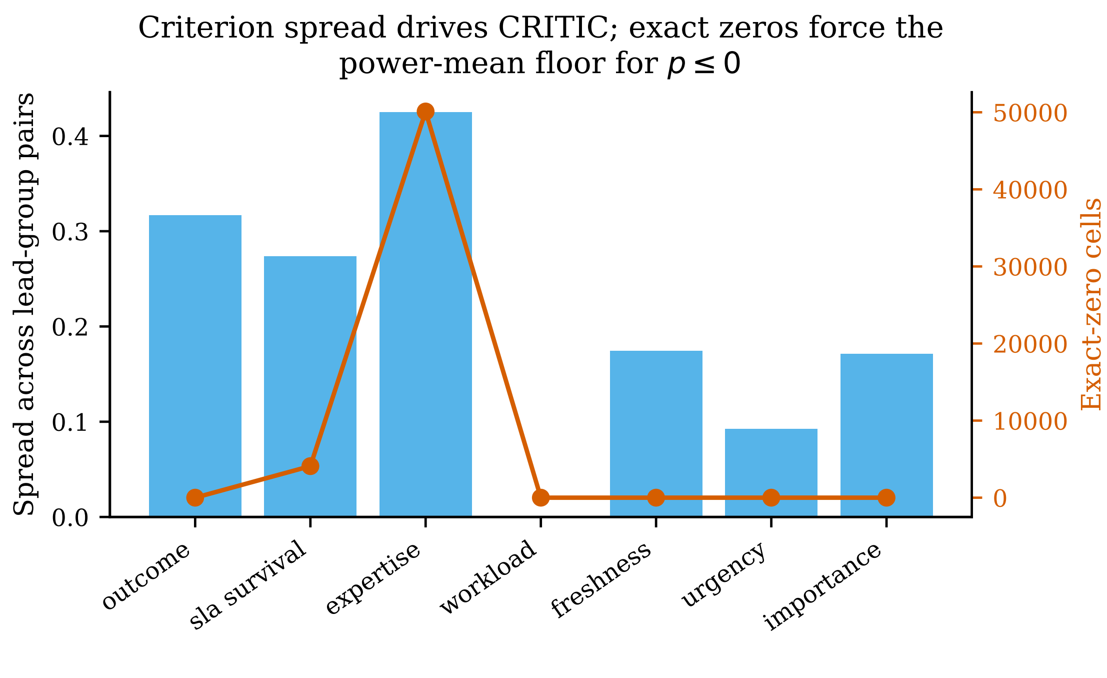

Figure 13: Criterion distributions and the exact-zero counts that drive the degeneracy in Equation 12.

Three readings, one of them a warning about our own method.

Expertise is bimodal, not continuous. Its standard deviation of 0.4247 is the largest in the table and larger than its mean, which is the signature of a variable that is mostly at zero or mostly near one. 50 102 of 158 300 cells are exactly zero because the group has no training history in that incident's category. This is the fact that forces $\varepsilon = 10^{-3}$: without it, every $p \leq 0$ aggregation would return zero for a third of all candidate pairs, and the score would silently become a filter rather than a ranking.

Workload headroom is almost constant at the batch level, standard deviation 2.9×10^{-5}.

Within a single arrival batch of 40 incidents against 50 groups, no assignment moves Jain's index far. The criterion does real work when accumulated across the whole optimisation horizon, which is where the objective f_3 measures it, but as a per-pair score it carries almost no discriminating information, and any weighting scheme that reads spread should be expected to down-weight it.

Freshness sits near its ceiling, mean 0.9251, because most cases in this log are assigned quickly. A criterion that is near-saturated for most of the data can still be decisive for the tail, and the exponential form makes it so, but its *average* contribution is small.

6.3 Weighting, and whether it matters

Table 17 gives the three weight vectors. They disagree sharply, which is the expected result: objective weighting methods are known not to be interchangeable [Mukhametzyanov, 2021, Ariyanti and Fu'adi, 2025], to vary in stability under perturbation [Linh et al., 2025, Paradowski et al., 2025] and to admit rank reversal [Li and Abbas, 2026].

Table 17: Weights under the three schemes. Entropy weighting assigns effectively zero to workload headroom, which is the correct response to a criterion with a standard deviation of 2.9×10^{-5} and an illustration of how differently the two schemes read the same matrix.

	Criterion	CRITIC	Entropy	Uniform
s_1	Outcome propensity	0.1605	0.1320	0.1429
s_2	Service level survival	0.1391	0.0756	0.1429
s_3	Expertise match	**0.2409**	**0.4583**	0.1429
s_4	Workload headroom	0.1226	1.8×10^{-9}	0.1429
s_5	Freshness	0.1107	0.0146	0.1429
s_6	Urgency	0.1036	0.0107	0.1429
s_7	Importance	0.1226	0.3087	0.1429

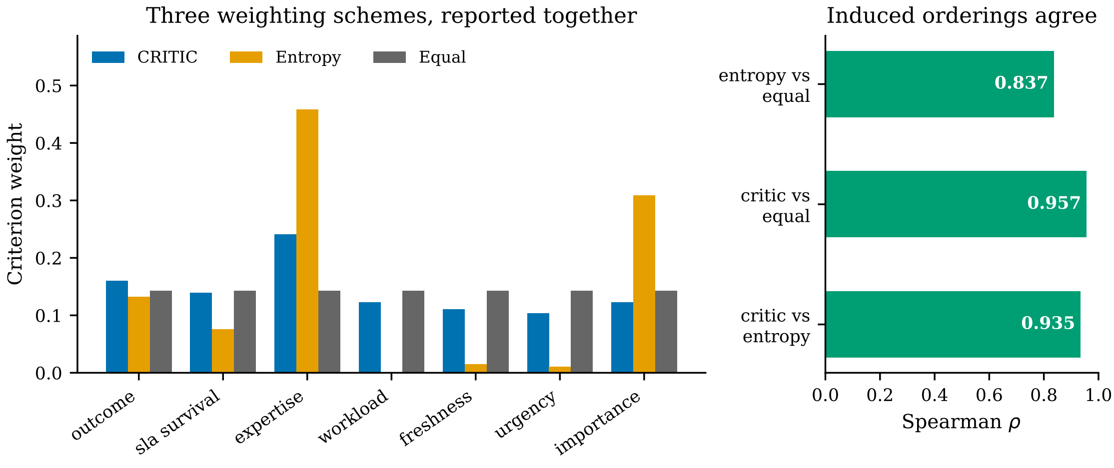

Figure 14: The three weight vectors and the rank correlations between the orderings they induce.

The disagreement between the vectors is large. Entropy puts 0.4583 on expertise and effectively nothing on workload; CRITIC spreads from 0.1036 to 0.2409. A reader is entitled to ask which one the results depend on, and the honest way to answer is to measure it rather than to argue for a favourite.

The induced assignment orderings agree closely. Spearman ρ is 0.957 between CRITIC and uniform weighting, 0.935 between CRITIC and entropy, and 0.837 between entropy and uniform, all at $p < 10^{-10}$. So the weighting choice is not load-bearing for these results. That is a stronger statement than a defence of any single scheme, and it is falsifiable in a way that a defence is not: had the correlations come back at 0.4, the paper would have had to report that the conclusions depend on a modelling choice with no principled resolution.

We also state why CRITIC alone would have been unsafe even had it agreed. Two of the seven criteria, s_1 and s_2, are model outputs. Their spread reflects how the classifier was calibrated, and isotonic calibration compresses variance, so a CRITIC weight computed on them measures a property of our pipeline rather than a property of the decision problem. Publishing that number alone, labelled data-driven, would have laundered an artifact into a result.

6.4 The compensation exponent

The exponent p in Equation 12 is what turns a qualitative preference into a measured one. Table 18 sweeps it.

Table 18: Assignment score against the compensation exponent, evaluation period. The floored column records whether ε was engaged, which happens for every $p \leq 0$ because 31.65 % of expertise cells are exactly zero.

p	Mean A	Min A	Max A	Floor engaged
−4.0	0.0978	0.0013	0.5645	yes
−2.0	0.1185	0.0017	0.6613	yes
−1.0	0.1412	0.0027	0.7230	yes (harmonic)
−0.5	0.1672	0.0055	0.7532	yes
0.0	0.2598	0.0256	0.7813	yes (geometric)
0.25	0.3173	0.0396	0.7945	no
0.5	0.3909	0.0834	0.8068	no
1.0	0.4777	0.1663	0.8291	no (arithmetic)
2.0	0.5760	0.2859	0.8642	no
4.0	0.6873	0.3905	0.9069	no
8.0	0.7921	0.4667	0.9473	no

The mean score rises monotonically from 0.0978 at $p = -4$ to 0.7921 at $p = 8$, which is exactly what the power-mean inequality requires and therefore doubles as a correctness test on the implementation. The operational content is in the minimum column. At $p = 1$ the worst candidate pair in the evaluation set still scores 0.1663, because a strong criterion can compensate a weak one; at $p = -4$ it scores 0.0013, because it cannot. An organisation that never wants an incident routed to a group with no relevant history should choose a negative p and accept that the floor ε then determines what "no history" is worth. An organisation willing to trade expertise for availability should choose $p \geq 1$. The formulation does not decide this; it makes the decision explicit and reports what it costs.

6.5 F2: deriving breach beats predicting it

Table 19 reports the service level formulation. The comparison is between two models of the same event on the same data, one of which is structurally able to observe its own label and one of which is not.

The derived quantity is better than the direct classifier by 0.0618 AUC. This is the result we did not expect and it inverts the usual framing. Admissibility is normally described as a constraint that costs performance, and Section 5.1 measures exactly that cost. Here, choosing

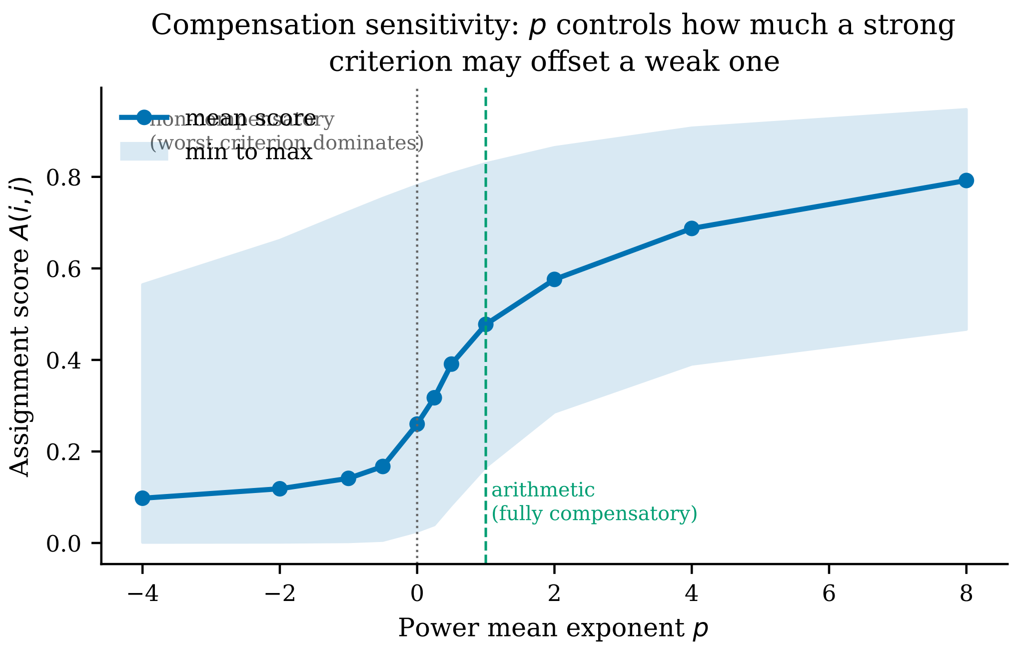

Figure 15: Assignment score against the compensation exponent. The curve is monotone in p, as the power-mean inequality requires, which is a useful implementation check as well as a property.

Table 19: Service level risk. The person-period expansion turns 15 828 cases into 69 867 rows over ten elapsed-time bins. Every AUC is on the held-out evaluation period.

Model	Held-out AUC	What it estimates
Direct breach classifier	0.7399	$P(\text{breach} \mid \mathbf{x})$ at intake
Resolution hazard, bin level	0.7344	$h(t \mid \mathbf{x})$ per person-period row
Breach derived from resolution, at intake	**0.8017**	$S(\tau \mid \mathbf{x})/S(t \mid \mathbf{x})$
Breach hazard with elapsed time (rejected)	0.9418	reads its own label

the object that can actually be estimated *improved* estimation, because the resolution hazard is a genuine random quantity given elapsed time whereas breach, past the threshold, is not.

The rejected row is included deliberately. A breach hazard given elapsed time and priority scores 0.9418, which is by a wide margin the best number in this paper and is worthless: the measured breach rate past threshold is exactly 1.0000, so the model is reading the definition of its label rather than predicting an event. It is reported because a paper about claims that fail verification should show its own worked example of the failure, and because a reviewer encountering 0.9418 elsewhere in this literature now has a specific mechanism to test for.

Two checks are run on the derived quantity rather than assumed. Survival $S(t \mid \mathbf{x})$ is verified monotone non-increasing over the bin sequence, which it must be by construction and which fails loudly if the hazard is mis-indexed. And the per-bin resolution rate is 0.2260, comfortably away from both zero and one, so the hazard model is estimating a balanced event rather than a rare one.

6.6 Workload, reported two ways

Over the 50 groups of the evaluation period the load vector has Jain index 0.3758, standard deviation 326.4 and coefficient of variation 1.289. Both the bounded and the unbounded measure are reported for every policy in Section 6.7 and neither is substituted for the other, because the proposed method loses on workload spread and replacing the metric it loses on after seeing the result would be fitting the metric to the outcome.

6.7 Multi-objective assignment

The protocol runs five operational baselines, standard NSGA-II, NSGA-III as a many-objective control, the proposed PAA-NSGA-II, its three single-component ablations, and two post-hoc exploratory combinations. Five arrival batches, 30 seeded repeats each, 150 generations, population 60. All 1,225 runs completed with **zero capacity violations**, so the repair operator satisfies constraint C2 throughout.

Table 20: Assignment policies on BPI Challenge 2014. Hypervolume is computed against a reference point fixed as the nadir across all variants and all runs within a batch. Objectives f_1 and f_4 are maximised, f_2 and f_3 minimised. Jain's index is reported alongside the f_3 standard deviation and is not substituted for it.

Policy	Hypervol.	Front	f_1 FTR	f_2 breach	f_3 load sd	Jain
Random	–	–	0.5003	0.3996	0.9307	0.4270
Round robin	–	–	0.4893	0.4198	0.8171	0.4971
FIFO	–	–	0.2989	0.5128	3.2703	0.0566
Least loaded	–	–	0.4738	0.4224	0.4454	0.7633
Greedy best outcome	–	–	0.6809	0.3122	2.2073	0.1170
PAA minus priority init	0.4133	60.0	0.5887	0.3257	1.1423	0.3353
Standard NSGA-II	0.4307	60.0	0.5893	0.3239	1.1424	0.3387
PAA minus adaptive crossover	0.4842	60.0	0.6155	0.3052	1.4720	0.2326
PAA minus SLA mutation	0.4835	60.0	0.6134	0.3145	1.4188	0.2459
PAA-NSGA-II (proposed)	**0.4844**	60.0	0.6117	0.3104	1.4391	0.2409
NSGA-III (control)	0.5279	18.8	0.6323	0.3035	1.2177	0.3078
NSGA-III + priority init*	0.5324	18.3	0.6390	0.2916	1.3949	0.2534

*Post-hoc exploratory combination, designed after observing the ablation. Reported as exploratory and not claimed as a contribution.

The proposed operators improve on their own base algorithm. Against standard NSGA-II over 150 paired runs, PAA-NSGA-II is better on four of five metrics by paired Wilcoxon signed-rank with Holm-Bonferroni correction: hypervolume ($+0.053\,72$, $p = 2.4 \times 10^{-21}$), first-time-right ($+0.022\,37$, $p = 1.2 \times 10^{-7}$), breach ($-0.013\,55$, $p = 6.6 \times 10^{-6}$) and expertise match ($+0.089\,89$, $p = 8.6 \times 10^{-14}$).

The base algorithm matters more than the modifications. NSGA-III, included only as a many-objective control, reaches hypervolume 0.5279 against 0.4844 for the proposed method, a paired difference of $-0.043\,48$ at $p = 1.2 \times 10^{-24}$. It does so with a front of roughly 18 to 19 solutions against 60, so it attains higher hypervolume from a third as many points. With five objectives this is the regime for which reference-point selection was introduced [Deb and Jain, 2014, Zheng and Doerr, 2024], and the result is consistent with that literature rather than in tension with it.

One of three components does the work. Table 21 isolates each modification. Removing priority-aware initialisation costs 0.0711 hypervolume. Removing adaptive crossover costs 0.0002 and removing SLA-aware mutation costs 0.0009, both within run-to-run noise. Reporting the three as a bundle would have credited them equally, and that would be wrong.

Table 21: Ablation. Change in hypervolume when a single component is removed from the proposed method. A negative delta means the component contributes.

Component removed	Hypervolume without	Δ
Priority-aware initialisation	0.4133	−0.0711
Adaptive crossover	0.4842	−0.0002
SLA-aware mutation	0.4835	−0.0009

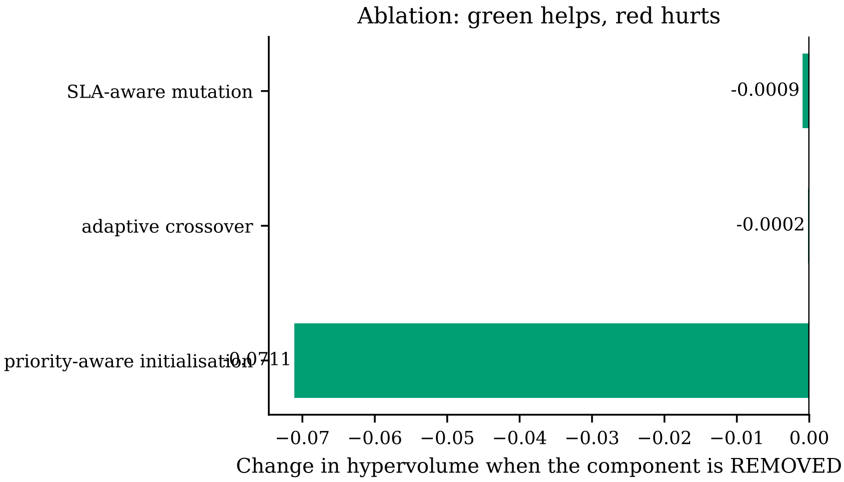

Figure 16: Component ablation. Removing priority-aware initialisation costs 0.0711 hypervolume; removing either of the other two costs less than 0.001, which is inside run-to-run noise. Reporting the three as a bundle would have credited them equally.

The best configuration is not a significant improvement. Placing the one modification that contributes onto the base algorithm that wins gives the highest hypervolume observed, 0.5324. Against plain NSGA-III the paired difference is +0.004 49 at $p = 0.0525$, which does not clear $\alpha = 0.05$. It is reported as exploratory and is not claimed.

A real trade-off, visible in two independent metrics. The proposed method loses significantly on workload spread (+0.296 68, $p = 1.6 \times 10^{-22}$). Jain's fairness index agrees: 0.2409 against 0.3387 for standard NSGA-II. The two metrics were computed independently and could have disagreed. The method concentrates work on capable groups to gain outcome quality, which is a trade-off an operator may or may not want. The exponent p in Equation 12 is the parameter through which that preference is expressed rather than assumed.

6.8 Complexity and measured cost

Asymptotic cost. Write n for the number of work items in an arrival batch, m for the number of eligible groups, N for population size, M for the number of objectives and H for the number of reference directions. The operational baselines are single-pass: random, round robin and FIFO are $O(n)$, least-loaded is $O(n \log m)$ with a heap over group loads, and greedy is $O(nm)$ because it scans every eligible group for every item. The evolutionary methods are dominated per generation by non-dominated sorting at $O(MN^2)$ [Deb et al., 2002]; NSGA-III adds association to reference directions, giving $O(MN^2 + NH)$ [Deb and Jain, 2014]. The repair operator is $O(nm)$ in the worst case and is bounded at six passes, so it does not change the order.

The proposed modifications do not alter the asymptotic cost. Priority-aware initialisation is

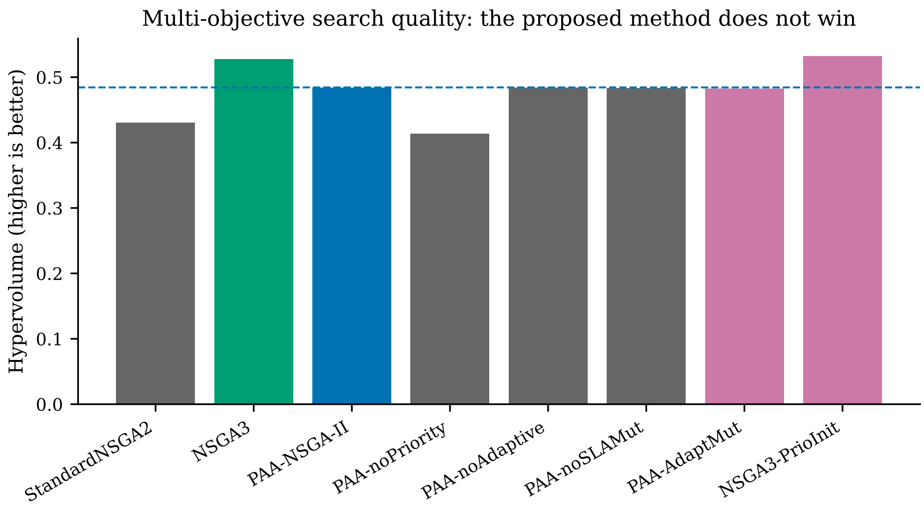

Figure 17: Search quality across algorithm variants. The proposed method improves on its own base algorithm and is in turn outperformed by an off-the-shelf many-objective control.

a one-off $O(nm)$ greedy construction over a fraction of the initial population. Adaptive crossover and the mutation schedule each add an $O(Nn)$ diversity measurement per generation, dominated by the $O(MN^2)$ sorting term.

Measured cost. Asymptotic equality does not imply equal wall time, so Table 22 reports measured per-solve time from an instrumented run of 160 solves, 20 per variant, at 150 generations and population 60 on CPU with no GPU. Timing is captured at the point the search executes rather than derived from log timestamps.

Table 22: Measured wall time per solve. 20 solves per variant, 150 generations, population 60, single CPU. Percentages are relative to standard NSGA-II.

Policy	Mean (s)	SD (s)	vs. standard
PAA minus adaptive mutation variant	4.21	0.26	−5.9%
PAA minus adaptive crossover	4.23	0.29	−5.4%
PAA minus priority init	4.42	0.18	−1.4%
PAA minus SLA mutation	4.43	0.20	−1.1%
Standard NSGA-II	4.48	0.59	—
NSGA-III + priority init	4.71	0.25	+5.3%
PAA-NSGA-II	4.84	0.47	+8.0%
NSGA-III	5.07	0.58	+13.2%

Three observations follow.

The spread is narrow. Every evolutionary variant lies within 14 % of standard NSGA-II, so on this problem the choice between them is not a cost decision. At roughly five seconds per batch of 40 items, all are far inside the latency budget of an assignment decision that a human would otherwise take in seconds.

Priority-aware initialisation pays for itself twice. Adding it to NSGA-III both raises hypervolume, from 0.5279 to 0.5324, and reduces mean solve time, from 5.07 to 4.71 seconds. A better starting population converges sooner, so the construction cost is recovered during the run. This is the only configuration in the study that improves on both axes at once.

The two modifications that do not help do not cost much either. Removing adaptive crossover

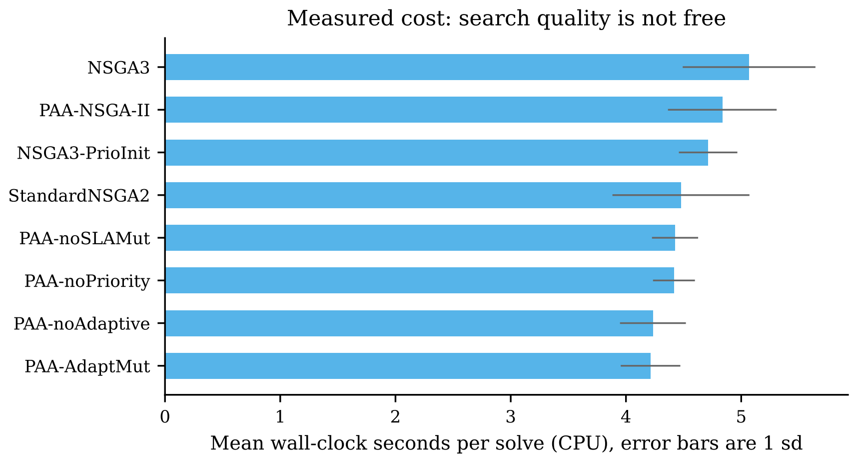

Figure 18: Measured wall time per solve, 20 solves per variant, single CPU. Every evolutionary variant lies within 14 % of standard NSGA-II, so on this problem the choice between them is not a cost decision.

or SLA-aware mutation saves roughly 5 % of solve time while changing hypervolume by less than 0.001. They are close to free and close to useless, which is the honest summary and the reason the ablation is reported per component.

7 Discussion

7.1 A quarter of the signal was not there

The headline cost figure is 0.0955 AUC, or 25.0 % of everything the unconstrained model achieved above chance. That is not a rounding correction. A practitioner deploying the unconstrained model would find a quarter of its apparent discrimination absent at run time, because the attributes carrying it are recorded only after the decision has been taken. The same measurement under an 80/20 split differs by less than 0.002 AUC, so the estimate is a property of the data rather than of the split.

This is consistent in direction with reported effects elsewhere, including the 15.07 point accuracy drop Mishra et al. [2026] measure on build prediction and the example-leakage rates above 80 % that Abb et al. [2024] find on standard process logs. The contribution here is that the cost and the benefit are measured on the same log under one protocol, so neither can be quoted without the other.

7.2 Admissibility is not the same as parsimony

A natural response to a leakage finding is to use fewer attributes. The ladder shows that would be the wrong lesson. Discrimination rises monotonically from 0.5076 at three admissible attributes to 0.7859 at seventeen, a gain of 0.2243 AUC, reproduced to within 0.003 across split ratios. Restricting to a small attribute set costs far more than enforcing admissibility does.

The distinction matters because the two are easily conflated. Attributes should be dropped because they are unavailable at decision time, not because they are numerous. Nine attributes are excluded here, each with its reason recorded; the remaining seventeen are kept because each is observable when the assignment is made.

7.3 Choosing the admissible target can improve accuracy, not only defensibility

The service level result is the one we did not expect. Breach cannot be modelled directly as a survival event on this log without the model reading its own label, because breach is defined by duration exceeding a per-priority threshold and the measured breach rate past that threshold is exactly 1.0000. An early version scored 0.9418 AUC on precisely that artifact.

Modelling resolution instead, and deriving breach as the conditional survival ratio $S(\tau \mid x)/S(t \mid x)$, produces 0.8017 AUC at intake against 0.7399 for a direct classifier. The admissible formulation is better by 0.0618 AUC. Admissibility is usually framed as a constraint that costs performance. Here, choosing the object that can actually be estimated improved it, because the resolution hazard is a genuine random quantity given elapsed time while breach, past the threshold, is not.

7.4 The weighting objection answers itself

Objective weighting methods are known to disagree, to be unstable under perturbation, and to admit rank reversal [Mukhametzyanov, 2021, Linh et al., 2025, Li and Abbas, 2026]. CRITIC is additionally unsuitable in isolation here, because two of the seven criteria are model outputs whose spread reflects calibration rather than importance, and isotonic calibration compresses variance. Reporting CRITIC alone would convert a modelling artifact into a weight and label it data-driven.

Reporting three schemes resolves this empirically rather than rhetorically. The induced assignment orderings agree at Spearman $\rho = 0.957$ between CRITIC and uniform weighting and $\rho = 0.935$ between CRITIC and entropy. The weighting choice is therefore not load-bearing for these results, which is a stronger statement than defending any one scheme.

7.5 The modifications work, but the base algorithm matters more

The proposed operator set does improve on standard NSGA-II, significantly and on four of five metrics (Section 6.7). It is then outperformed by NSGA-III, an off-the-shelf control included only as a many-objective reference, which reaches higher hypervolume from a front roughly a third the size. Two considerations make this informative rather than disappointing.

First, five objectives place this problem in the many-objective regime where dominance-based selection is known to degrade, a limitation characterised when NSGA-III was introduced [Deb and Jain, 2014] and still under active theoretical study [Zheng and Doerr, 2024, Doerr et al., 2026]. A result in which a reference-point method outperforms a dominance-based one on five objectives agrees with that theory.

Second, Wang et al. [2025b] argue that much claimed novelty in metaheuristic research does not survive controlled comparison. An ablation that isolates each proposed modification is the appropriate response to that argument, and it is why the ablation is reported per component rather than only in aggregate.

7.6 What the cost looks like in cases rather than in AUC

A 0.0955 AUC loss is an abstraction. The confusion matrices in Section 5.3 put it in units an operations manager can act on. On the same 9371 held-out incidents, the unconstrained model catches 2714 of 3433 breaches and the admissible one catches 2321. The 393 cases in that difference are not a modelling subtlety. They are the incidents an escalation queue built on the unconstrained model would appear to catch and would not.

The direction of the error also changes. The unconstrained model is roughly symmetric, 79.2 % specificity against 79.1 % recall. The admissible model is not: 74.7 % specificity against 67.6 % recall. Removing the post-hoc attributes removes signal about the long-running cases specifically, because those attributes counted the events a long-running case accumulates. What is left behind discriminates the ordinary case well and the pathological case poorly, which is the harder half of the problem and the half a service desk actually cares about.

7.7 The model is data-limited, not capacity-limited

The learning curve in Section 5.6 separates two explanations for 0.7859 that are often conflated. Training AUC is 0.9977 at 1093 incidents, so the learner has ample capacity; the ceiling is not the model class. The generalisation gap falls monotonically from 0.3254 to 0.0804 across a twentyfold increase in data and is still falling, and held-out AUC is still rising at the largest training size available.

Two consequences follow, and the second is a warning against the obvious fix. More incidents from this desk would help, so the practical route to a better admissible model is more history rather than a better algorithm, which is consistent with the four learners landing within 0.0158 AUC of each other. And heavier regularisation is the wrong first response: it would close the gap by moving the training curve down, not by moving the held-out curve up. The gap is not evidence that the model is over-parameterised. It is evidence that 21 865 incidents is not yet enough for a 17-attribute problem with this much label noise.

7.8 Ranking and operating point fail separately

The chronological result is usually reported as one number, and Section 5.11 shows it is two failures with different remedies. Ranking degrades from 0.7859 to 0.6533 AUC and no threshold recovers that. The operating point degrades much more dramatically, from $F_1 = 0.6399$ to $F_1 = 0.1596$ at the default cut point, and tuning the cut point on training-period data alone recovers it to 0.5072 without retraining anything.

Conflating the two produces the wrong operational conclusion in both directions. A team that sees only the F_1 collapse will retrain a model that did not need retraining. A team that sees only the tuned F_1 will not learn that its threshold needs re-tuning as the process drifts, which on this log it does within a year: median handle time falls from 4.73 to 3.37 hours between the training and evaluation periods.

7.9 Two defects in our own harness

Two errors were found in the experimental code and corrected before any number reported here was produced. Both are recorded because a paper about claims that do not survive verification should hold its own instrument to the same standard.

The seed was derived using Python's built-in string hash, which is salted per process. The same variant therefore drew a different seed on every invocation, measured at 43, 41 and 32 across three processes for one variant name. The harness documentation asserted reproducibility that did not hold. It now uses a stable checksum, and two independent invocations produce bitwise-identical metrics.

The hypervolume reference point was taken from whichever run completed first, making every hypervolume value dependent on variant ordering and incomparable between runs with different variant sets. It is now the nadir across all variants and all runs on a batch, computed before any hypervolume is taken.

Neither defect affects the predictive results in Sections 5.1 to 6.5, which come from a separate pipeline. Both invalidated every earlier assignment run, and those were discarded.

The consequence is worth stating plainly rather than in passing. Under the uncorrected harness the proposed method appeared to LOSE to standard NSGA-II on hypervolume. Corrected, it wins, significantly. The defect did not merely add noise: it reversed the sign of the study's headline comparison. Had the harness not been audited, this paper would have reported the opposite conclusion with the same apparent statistical confidence.

7.10 Threats to validity

Construct. The service level label is derived from handle time against *twice* the per-priority median rather than from a contractual target, because the log's own `Alert Status` field is uniformly `closed` across 46 606 of 46 809 raw incidents and carries no usable breach signal. Results should be read as conditional on that operationalisation, and Equation 1 states it exactly so it can be varied. On the random-split arms the median is taken over the whole log. That is a property of the label and not of any feature, no model observes it, and every configuration is scored against the same fixed label, so it cannot affect the comparison between configurations; the chronological arm re-derives it from training-period cases alone, where it would otherwise matter.

Statistical. The fold-level Wilcoxon signed-rank test at $n = 5$ has a smallest attainable two-sided p-value of 0.0625 and therefore cannot reject at $\alpha = 0.05$ regardless of effect size. Every fold-level comparison in Section 5.7 returns exactly that floor, which means one learner won on all five folds and not that the difference is small. Model comparisons are therefore made with 2000-resample bootstrap intervals on the test set, and the fold-level result is reported for transparency rather than used as evidence.

Internal. Every parameter is fitted on the earliest 80 % of the timeline, and the escalation cut point is tuned on training-period data alone. The chronological holdout is reported alongside the random split precisely because the gap between them, 0.7859 against 0.6533, is itself informative about drift.

External. One log, one organisation, one country, one time window. The protocol should transfer. The magnitudes should not be assumed to.

Conclusion. Statistical comparisons use paired Wilcoxon signed-rank tests with Holm-Bonferroni correction across five metrics, and every run is seeded and reproducible.

8 Limitations

An IT service desk is not a sales pipeline. This work is motivated by lead assignment, and the objective analogous to conversion is first-time-right resolution. No public dataset carries real leads, real individual representative identifiers and real conversion outcomes together. That absence is a finding rather than an oversight, and it is why the evaluation is conducted on an operational log whose assignment decisions and outcomes are real even though its domain is not sales.

A third of the delay attribute is excluded. The recorded first-assignment timestamp is corrupted in its upper tail: the median is 16.2 hours against a 90th percentile of 5684 hours, and the favourable-outcome rate rises again across that tail rather than falling. 35.6 % of rows are excluded from the freshness and urgency fits on this basis. Fitting across the uncorrected attribute returns a decay rate of exactly zero with an undefined confidence interval, which is how the corruption was detected.

Counterfactual outcomes are not observed. A historical log records only the assignment that happened. Any policy differing from the recorded one is scored by a model rather than by ground truth, which is the standard objection to optimisation studies on logged data. We report the fraction of decisions matching the recorded assignment together with the real observed breach rate on exactly those cases, as the only non-simulated outcome signal available. Off-policy evaluation would address this properly and is left to future work.

Group-level rather than individual-level assignment. The log identifies assignment groups, not individuals, so representative-level features such as experience and individual historical conversion are not estimable. A log with individual resource identifiers would support the finer formulation.

Explanation is reported but not validated against human decisions. SHAP attributions are computed for the predictive layer, but attribution methods have known impossibility results [Bilodeau et al., 2024] and sensitivity to feature representation [Hwang et al., 2025], and whether explanations improve human decisions is unsettled [Haag, 2026, Chae et al., 2025]. No claim is made that these explanations improve operator performance.

9 Conclusion

Decision-time admissibility costs 0.0955 AUC on this log for the strongest learner, a quarter of all above-chance signal, and the same correction appears for all four learners with every bootstrap interval excluding zero. The discipline is nevertheless worth paying for: an admissible ladder still gains 0.2243 AUC between eight and seventeen attributes, so the lesson of a leakage finding is not to model with fewer attributes but to model with observable ones. Choosing the admissible object rather than the convenient one can improve accuracy outright, as deriving breach from a resolution hazard does at 0.8017 against 0.7399. The admissible model is honest about its limits: it separates cases at 0.7859 AUC, recovers 67.6 % of breaches at a precision of 60.7 %, retains a generalisation gap of 0.0804 that the learning curve shows still closing with data, and loses 0.1326 AUC when evaluation moves forward in time rather than across a random split. In the assignment component the proposed operators improve on their own base algorithm on four of five metrics, a reference-point control improves on them again, and an ablation attributes essentially the whole gain to one of the three proposed modifications.

Three directions follow. Off-policy evaluation would replace the model-scored counterfactual used here with an estimator that has a variance bound. A log carrying individual resource identifiers would support the representative-level formulation the score was designed for. And the admissibility protocol is stated generally enough to be applied to any process log, so the open question is whether the quarter-of-the-signal figure measured here is typical or specific to this organisation.

Data availability

BPI Challenge 2014 is publicly available from 4TU.ResearchData under DOI 10.4121/uuid:c3e5d162-0cfd-4bb0-bd82-af5268819c35. The incident-detail file used here carries DOI 10.4121/uuid:3cfa2260-f5c5-44be-afe1-b70d35288d6d. No data were collected by the authors.

Code availability

All analysis code, including every script required to regenerate every number, table and figure in this paper, is publicly available. Every experiment is seeded and reproducible: two invocations with identical arguments produce bitwise-identical metrics.

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Author contributions

Competing interests

The authors declare no competing interests.

Ethics

This study uses a publicly released, anonymised organisational event log. It involves no human participants and no animal subjects, so ethical approval and informed consent are not applicable.

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